

Two Certified Bounds on the Universal i.i.d. Berry–Esseen Constant:

$$0.40 \leq C_{\text{iid}} \leq 0.4688$$

A formalization report on the **BerryEsseen** Lean development

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Abstract

Let $X_1, X_2, \dots$ be independent, identically distributed real random variables with $\mathbb{E}X_1 = 0$, $\mathbb{E}X_1^2 = 1$ and $\beta := \mathbb{E}|X_1|^3 < \infty$, let F_n be the distribution function of $(X_1 + \dots + X_n)/\sqrt{n}$, and let Φ be the standard normal distribution function. The universal i.i.d. Berry–Esseen constant C_{iid} is the least constant C for which $\sup_x |F_n(x) - \Phi(x)| \leq C\beta/\sqrt{n}$ holds for every such law and every $n \geq 1$. This document presents, in ordinary mathematical language, the complete proofs of the two bounds

$$0.40 \leq C_{\text{iid}} \leq 0.4688 = \frac{293}{625},$$

as they are formalized in the Lean 4 development **BerryEsseen** on top of **mathlib**. The two statements are exactly the two declarations that the project’s pinned, sandboxed comparator is configured to check.

The lower bound comes from an explicit six-summand standardized Bernoulli witness with success probability $2/5$, evaluated by exact rational arithmetic, together with one rational estimate of a normal probability. For completeness we also give the (not yet formalized) asymptotic proof of Esseen’s stronger lower bound $(\sqrt{10} + 3)/(6\sqrt{2\pi})$.

The upper bound is a Fourier-analytic argument that removes the unknown law analytically and then verifies interval inequalities. Its ingredients are: cosine-anchor estimates that retain the information that $\beta = 1$ forces the symmetric two-point law; a global exponential modulus bound obtained from symmetrization and an explicitly constructed convex minorant; Prawitz’s smoothing inequality, proved from scratch through a shift recurrence and the Riemann–Lebesgue lemma; a power-comparison step that keeps the Gaussian damping inside a geometric average; and four exhaustive parameter regions — an analytic small-fraction region with the stronger regional constant $183/400$, an elementary Cantelli region, 467 certified interval cells covering all sample sizes $n \geq 20$ or third moments $\beta \geq 2$, and 174 certified finite-sample cells covering $1 \leq n \leq 19$, $\beta < 2$.

All finite certificates are checked in Lean’s kernel by **decide +kernel**; the development contains no **sorry** and no **native_decide**, and its axiom closure is **{propext, Classical.choice, Quot.sound}**. The final pinned sandboxed comparator run is *pending* at the time of writing; nothing in this document should be read as reporting a completed comparator verification.

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1 Introduction

1.1 The constant

Let X be a real random variable with

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \beta := \mathbb{E}|X|^3 < \infty, \quad (1)$$

let $X_1, X_2, \dots$ be independent copies of X , and put

$$S_n := \frac{X_1 + \dots + X_n}{\sqrt{n}}, \quad F_n(x) := \mathbb{P}(S_n \leq x), \quad \Delta_n := \sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)|.$$

The Berry–Esseen theorem [1, 2] asserts the existence of an absolute constant C with $\Delta_n \leq C\beta/\sqrt{n}$ for all $n \geq 1$ and all laws satisfying (1). We write C_{iid} for the least such constant and

$$\ell := \frac{\beta}{\sqrt{n}}$$

for the *Lyapunov fraction*, so that the assertion to be proved is $\Delta_n \leq C_{\text{iid}} \ell$ and $C_{\text{iid}} = \sup \Delta_n / \ell$, the supremum being over all admissible laws and all $n \geq 1$.

Table 1 records the published history of two-sided bounds for C_{iid} , following the survey in [12]. Zolotarev’s conjecture [17] is that C_{iid} equals Esseen’s lower bound

$$c_E := \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = 0.4097321837\dots,$$

and this is known in the i.i.d. Bernoulli (binomial) subcase [7, 18].

Bound	Reference	Comment
$C_{\text{iid}} \leq 0.82$	[17]	Zolotarev-type smoothing inequalities; [17] also gives 0.9051 for the non-i.i.d. case.
$C_{\text{iid}} \leq 0.7975$	[15]	Fourier-analytic refinement.
$C_{\text{iid}} \leq 0.7655$	[11]	
$C_{\text{iid}} \leq 0.7056$	[8]	
$C_{\text{iid}} \leq 0.5129$	[4]	From the structural bound $\Delta_n \leq 0.34445(\beta + 0.489)/\sqrt{n}$.
$C_{\text{iid}} \leq 0.4785$	[14]	
$C_{\text{iid}} \leq 0.4748$	[10]	
$C_{\text{iid}} \leq 0.4690$	[9]	From $\Delta_n \leq n^{-1/2} \min\{0.4690\beta, 0.3322(\beta + 0.429), 0.3031(\beta + 0.646)\}$.
$C_{\text{iid}} \geq c_E$	[3]	$c_E = (\sqrt{10} + 3)/(6\sqrt{2\pi}) \approx 0.4097321837$, attained asymptotically by normalized Bernoulli sums.

Table 1: Published bounds for the universal i.i.d. Berry–Esseen constant, as collected in [12].

1.2 The two theorems of this document

The Lean development `BerryEsseen` proves two separate statements about the same defined object. In Lean they read as follows (the extended nonnegative reals are written `ENNReal` here; the source file uses the Unicode notation for that type):

```
theorem berry_esseen_constant_lower_bound :
  (0.40 : ENNReal) <= BerryEsseen.berryEsseenConstant
```

```
theorem berry_esseen_constant_upper_bound :
  BerryEsseen.berryEsseenConstant <= 0.4688
```

and the object `BerryEsseen.berryEsseenConstant` is defined as an extended nonnegative real supremum, so that finiteness of C_{iid} is a consequence of the second theorem rather than a hypothesis. The precise definitions are reproduced in Section 2.

Theorem 1.1 (Lower bound; `BerryEsseen.berryEsseenConstant_lower_bound`). $\frac{2}{5} \leq C_{\text{iid}}$.

Theorem 1.2 (Upper bound; `BerryEsseen.berryEsseenConstant_upper_bound`). $C_{\text{iid}} \leq \frac{293}{625} = 0.4688$.

These two declarations are exactly the *comparator targets*: the file `comparator/config.json` lists `berry_esseen_constant_lower_bound` and `berry_esseen_constant_upper_bound` as the two theorem names to be compared against the trusted challenge file, with permitted axiom list `{propext, Quot.sound, Classical.choice}`.

1.3 Shape of the upper-bound proof

The obstruction to a short proof of a numerically good constant is that the unknown law enters through its characteristic function $f(t) = \mathbb{E}e^{itX}$, and that the Kolmogorov distance is sensitive to the whole frequency range, including lattice laws whose characteristic functions return to modulus one. The proof organizes this as follows.

1. **anchors** (Section 4). The third absolute moment satisfies $\beta \geq 1$ with equality only for the symmetric two-point law, whose characteristic function is $\cos t$. Quantitatively $\mathbb{E}[(|X| - 1)^2 (2|X| + 1)] = 2(\beta - 1)$, and this identity yields

$$|\operatorname{Re} f(t) - \cos t| \leq \frac{\beta-1}{6}|t|^3, \quad |\operatorname{Im} f(t)| \leq \min\left\{\frac{|t|^3}{6}\sqrt{(\beta-1)(\beta+\frac{5}{3})}, \frac{t^2}{2}\sqrt{\beta-1}, \frac{\beta|t|^3}{6}\right\}.$$

Both bounds degenerate as $\beta \downarrow 1$, which the classical third-order Taylor estimate does not.

2. **Global modulus** (Section 5). Symmetrization and an explicitly constructed convex, nonincreasing minorant h of $s \mapsto (1 - \cos s)/s^2$ give $|f(t)| \leq \exp\{-t^2 h((\beta+1)|t|)\}$ at *every* frequency.
3. **Prawitz smoothing** (Section 6). A pointwise majorant of the indicator of $[0, \infty)$, built from a truncated cotangent Fourier kernel, converts the two characteristic-function estimates into four positive real integrals over a bounded frequency band plus one explicit Gaussian tail.
4. **Power comparison** (Section 7). Because $f_n(t) = f(t/\sqrt{n})^n$, a difference of n -th powers must be estimated. Keeping the differing decay rates of the two bases inside a geometric average, rather than bounding the geometric sum by n times its largest term, is what makes a single inequality valid for infinitely many sample sizes.
5. **Four exhaustive regions** (Sections 8–11). See Table 2.

Region	Method	Section	Regional bound
$\ell \leq \frac{1}{20}$	analytic, no computation	8	$\frac{183}{400} = 0.4575$
$\ell \geq \frac{6}{5}$	Cantelli, no computation	9	$\frac{11}{24} = 0.458\bar{3}$
$\ell \geq \frac{1}{20}$ and $(\beta \geq 2 \text{ or } n \geq 20)$	467 certified interval cells	10	$\frac{293}{625} = 0.4688$
$1 \leq n \leq 19, 1 \leq \beta < 2, \ell < \frac{6}{5}$	174 certified finite cells	11	$\frac{293}{625} = 0.4688$

Table 2: The four parameter regions of the upper-bound proof. The union of the regions is the whole admissible parameter set; see Section 13.

1.4 What is proved, and what is verified

It is important to keep three different claims apart.

1. *Mathematics*. Theorems 1.1 and 1.2 are formalized in Lean 4 with `mathlib`. There is no `sorry` in the development, and no published Berry–Esseen bound is imported as an assumption. The axiom closure of the two theorems is `{propext, Classical.choice, Quot.sound}` — the standard `mathlib` axioms — and in particular `native_decide` is *not* used anywhere in this development.

2. *Finite checks.* All finite certificates are discharged by `decide +kernel`, i.e. they are evaluated by Lean’s kernel on exact integer and rational arithmetic. A direct count over the certificate directories gives 961,710 occurrences of `decide +kernel` (Table 7).
3. *Comparator.* The project’s intended final gate is a fresh, sandboxed Linux run of the pinned `leanprover/comparator` [5], which re-elaborates the statement against the self-contained challenge file, checks the axiom whitelist, and replays the dependency closure in Lean’s kernel. **That run is pending.** Neither the repository nor this document claims comparator completion. A passing ordinary `lake build` is a different and weaker event, and is reported separately; see Section 14.

1.5 A second, sharper development in the same repository

The same repository contains a separate development that proves the sharper bound $C_{\text{iid}} \leq 0.423$ (`BerryEsseen.Upper0423.berryEsseenConstant_le_0423` in `BerryEsseen/Upper0423/Main.lean`, documented in `docs/UPPER_0423.md`). That development uses `native_decide` for its finite certificates, so its axiom report contains, besides the three standard axioms, one auxiliary axiom per finite certificate (579 in total); it therefore trusts Lean’s compiler and evaluator in addition to the kernel, a strictly weaker trust boundary than the one described in Section 1.4. It does not modify, and is not imported by, the 0.4688 development: the default library target does not import `Upper0423`, so the comparator configuration is unaffected. The two theorems presented in this document, and not the 0.423 theorem, are the comparator targets.

1.6 Provenance and attribution

Because the same repository hosts two developments with different provenance, we state explicitly what the repository’s own documentation records.

The 0.4688 development discussed in this document contains *no* statement attributing any of its ingredients to an external source beyond the classical literature. A search of `docs/` and of the Lean sources outside `BerryEsseen/Upper0423/` for attribution language (“due to”, “adapted from”, “based on”, “attribution”, author names) returns only the names of classical results that the proofs invoke and prove for themselves: Prawitz’s smoothing inequality and his kernel [6]; the one-sided Chebyshev (Cantelli) inequality; Jordan’s sine inequality; the Riemann–Lebesgue lemma; Taylor’s theorem; Cauchy–Schwarz; and Esseen’s lower-bound construction [3], whose Bernoulli coefficient the notes take from the historical record in [18]. The broader strategy — Fourier smoothing with a finite frequency band, a moment-dependent family of characteristic-function estimates, and a split of the parameter space into regions treated by different arguments — is the strategy of the literature summarized in Table 1, in particular of [17, 15, 11, 8, 4, 14, 10, 9].

The one attribution section in the repository is in `docs/UPPER_0423.md`, and it concerns the *separate* 0.423 development of Section 1.5. It reads as follows (its bullet list is run together here, and its per-file cross-references are omitted; the ellipsis marks the omission of a numerical bracketing of κ and of the coarser rational majorant used there):

“The analytic framework of this proof is not new here. The research notes `research/imaginary-envelope.md` (“Source alignment and attribution”) and `research/recovery/RECOVERY.md` attribute the following ingredients to [haonan-xiao/iid-berry-esseen, commit f00d781c], which proves a 0.4395 bound for the same constant: the shared berry parameters, the radial rectangle and the disk/strip discussion, the shared convex profile, the moment region, and the Prawitz normalization convention; the two-point sine-circle mechanism; the stop-loss identity for $\mathbb{E}|X - X'|^3$ and the resulting bounds on the symmetrization excess η ; the separate radial estimates for the real and imaginary parts; the cosine constant $\kappa = \max_{v>0}(\cos v - 1 + v^2/2)/v^3$... and the same convex profile $1/2 - \kappa v$ continued by $(1 - \cos v)/v^2$.”

The cited research notes are not part of the repository; on the machine used to prepare this document they sit in a sibling `research/` directory next to the checkout. They record, conversely, that two of the ingredients used by the 0.423 argument are taken *from* the present 0.4688 development: they list “our restored repository’s `BerryEsseen/WeightedMoments.lean`, `ImaginaryPart.lean`, and `TrigonometricAnchors.lean`” as the source of “the earlier beta-minus-one anchor estimates”, and state that “the restored project’s `docs/ANCHORS.md` and `docs/PROOF_OVERVIEW.md` retain the original anchor and geometric-sum arguments”.

Two of the ingredients that `docs/UPPER_0423.md` attributes to [16] do have counterparts in the 0.4688 development described here, and we record this as a factual overlap without any claim about priority:

1. the *convex cosine profile*: the minorant h of Section 5 has exactly the shape “a linear branch $\frac{1}{2} - cv$ continued by a scaled $(1 - \cos v)/v^2$ ”. The present development uses the coarser rational slope $c = \frac{1}{10}$, justified by its own lemma $\cos v - 1 + v^2/2 \leq |v|^3/10$ (`BerryEsseen.cosine_tenth` in `BerryEsseen/CosineMinorant.lean`), and splices at $v = 9/2$;
2. the *Prawitz normalization convention*: the present development uses the doubled positive-frequency kernel $K_\sigma(s) = \sigma(1 - s) + iB(s)$ with no compensating factor $\frac{1}{2}$ outside the integral, and says so explicitly in `docs/PRAWITZ_MAJORANT.md` and `docs/SMALL_FRACTION.md` (Convention 1.3 below).

Beyond this, the anchors of Section 4 and the geometric-sum arguments of Section 7 are the arguments that the research notes ascribe to this development itself. This document makes no assessment of anyone’s priority or of the correctness of [16].

1.7 Conventions and reading guide

Throughout, Φ and φ are the standard normal distribution function and density, $f(t) = \mathbb{E}e^{itX}$, $g(t) = e^{-t^2/2}$, $\text{sinc}(u) = \sin(u)/u$ for $u \neq 0$ and $\text{sinc}(0) = 1$, and

$$q(x) := \frac{1}{x} - \cot x \quad (0 < x < \pi).$$

All named constants are exact rationals or explicit algebraic numbers; no floating-point number occurs in any statement.

Convention 1.3 (Kernel normalization). For $\sigma \in \{+1, -1\}$ and $0 < s < 1$ set

$$B(s) := (1 - s) \cot(\pi s) + \frac{1}{\pi}, \quad K_\sigma(s) := \sigma(1 - s) + iB(s), \quad K := K_{+1}.$$

This K is *twice* the kernel that is usually called the Prawitz kernel; the compensating factor 2 that normally stands outside the integral is absent. A reader comparing formulas with a source that uses the half-normalized kernel must insert factors of two in the first three of the four smoothing terms (the tail term is unchanged). The Lean definition is `BerryEsseen.prawitzKernel` in `BerryEsseen/PrawitzFourier.lean`.

Each mathematical statement below is followed by a LEAN note giving the name of the corresponding declaration and its file, so that a reader can locate the formal counterpart of every step. Where a proof is long or purely computational we give a faithful sketch and the pointer; where a proof is short we give it.

2 The formal statement

The definitions below are reproduced from `BerryEsseen/Defs.lean`; the comparator’s trusted file `Challenge.lean` repeats them verbatim and imports only `mathlib` [13], so that nothing about the meaning of the statement depends on the project’s own modules. In particular the normal law is `mathlib’s ProbabilityTheory.gaussianReal 0 1`, the product measure is `mathlib’s MeasureTheory.Measure.pi`, and integrability, measurability and the supremum are `mathlib’s`.

2.1 Admissible laws

Definition 2.1. For a Borel probability measure μ on $\mathbb{R}$ put

$$\beta(\mu) := \int_{\mathbb{R}} |x|^3 d\mu(x) \in [0, \infty].$$

The measure μ is *admissible* if

- (a) $x \mapsto x$, $x \mapsto x^2$ and $x \mapsto |x|^3$ are μ -integrable;
- (b) $\int x d\mu = 0$;
- (c) $\int x^2 d\mu = 1$.

In Lean, $\beta(\mu)$ is `BerryEsseen.thirdAbsMoment` and Definition 2.1 is the structure `BerryEsseen.Admissible`, whose five fields are exactly (a)–(c). Note that integrability is part of the definition rather than an afterthought: the class of laws is not restricted to bounded or discrete laws, and no moment beyond the third is assumed.

2.2 Normalized sums via a product measure

Independence is realized by a finite product measure, so that the object whose distribution function is estimated is literally the law of a sum of n independent copies.

Definition 2.2. For $n \in \mathbb{N}$ define $\Sigma_n : \mathbb{R}^{\text{Fin } n} \rightarrow \mathbb{R}$ by $\Sigma_n(x) = (\sum_i x_i)/\sqrt{n}$, and let

$$\mu_n := (\Sigma_n)_* \left(\bigotimes_{i=1}^n \mu \right)$$

be the pushforward of the n -fold product measure. Let $\Phi(x) := \gamma((-\infty, x])$ where γ is `mathlib's ProbabilityTheory.gaussianReal 0 1`.

LEAN. `normalizedSum, normalizedSumLaw, normalCDF` in `BerryEsseen/Defs.lean`.

2.3 The constant as a supremum

Definition 2.3. For admissible μ , $n \geq 1$ and $x \in \mathbb{R}$ put

$$N(\mu, n, x) := \left(\frac{\sqrt{n}}{\beta(\mu)} |\mu_n((-\infty, x]) - \Phi(x)| \right)_+ \in [0, \infty],$$

the value being taken in the extended nonnegative reals, and define

$$C_{\text{iid}} := \sup_{\mu \text{ admissible}} \sup_{n \geq 1} \sup_{x \in \mathbb{R}} N(\mu, n, x) \in [0, \infty].$$

LEAN. `normalizedError` and `berryEsseenConstant` in `BerryEsseen/Defs.lean`. In Lean, C_{iid} is an `ENNReal`-valued iterated supremum $\bigsqcup_{\mu} \bigsqcup_{h: \text{Admissible } \mu} \bigsqcup_{n: 0 < n} \bigsqcup_x \text{normalizedError } \mu \ n \ x$, and $N(\mu, n, x) = \text{ENNReal.ofReal}(\dots)$.

Three features of Definition 2.3 deserve emphasis, since they are what make the upper bound a substantive statement.

1. The value is *a priori* allowed to be $+\infty$. Finiteness of C_{iid} is therefore a theorem (a consequence of Theorem 1.2) and not a side condition imposed on the definition.
2. The supremum is over *all* admissible laws. There is no restriction to a finite family, to laws with bounded support, or to numerically sampled distributions.
3. The inner supremum over x is over all real thresholds, and $\mu_n((-\infty, x])$ is a genuine right-continuous distribution function; atoms are permitted and are handled explicitly in the smoothing step (Section 6).

2.4 The reduction used by both bounds

The following triviality is the interface between Definition 2.3 and everything that follows.

Lemma 2.4. For every $C \in [0, \infty]$,

$$C_{\text{iid}} \leq C \iff (\forall \mu \text{ admissible}, \forall n \geq 1, \forall x \in \mathbb{R} : N(\mu, n, x) \leq C).$$

Moreover $N(\mu, n, x) \leq C_{\text{iid}}$ for every admissible μ , $n \geq 1$, x .

Proof. Both directions are the universal property of an iterated supremum in the complete lattice $[0, \infty]$. $\square$

LEAN. `BerryEsseen.constant_le_iff` and `BerryEsseen.normalizedError_le_constant` in `BerryEsseen/Constant.lean`.

Thus the lower bound is proved by exhibiting one triple (μ, n, x) , and the upper bound is proved by a pointwise estimate valid for all triples. The translation from a bound on the Kolmogorov distance to a bound on $N(\mu, n, x)$ is the following elementary step, used by every region of the upper-bound proof.

Lemma 2.5. Let μ be admissible, $n \geq 1$, $C \geq 0$, $a \in \mathbb{R}$ with $a \leq \beta(\mu)/\sqrt{n}$, and suppose $|\mu_n((-\infty, x]) - \Phi(x)| \leq C a$. Then $N(\mu, n, x) \leq C$.

Proof. Since $\beta(\mu) > 0$ for admissible μ (Lemma 4.1), multiply the hypothesis by $\sqrt{n}/\beta(\mu) > 0$ and use $a \leq \beta(\mu)/\sqrt{n}$ to get $\frac{\sqrt{n}}{\beta} |\dots| \leq \frac{\sqrt{n}}{\beta} C \frac{\beta}{\sqrt{n}} = C$. $\square$

LEAN. `BerryEsseen.normalizedError_le_of_cdf_bound` in `BerryEsseen/ParameterCellBounds.lean`; the variant used by the finite-sample cells, which replaces $\sqrt{n}/\beta$ by $\sqrt{H}/L$ for a cell with $n \leq H$ and $L \leq \beta$, is `BerryEsseen.normalizedError_le_finite_budget` in `BerryEsseen/FiniteSmoothing.lean`.

The asymmetry between a and $\beta/\sqrt{n}$ in Lemma 2.5 is the mechanism by which finitely many cells cover a continuum of parameters: on a cell whose Lyapunov fraction ranges over $[a, b]$, the analytic majorants are evaluated at the *upper* endpoint b (so that they dominate every law in the cell) while the numerical budget is compared against the *lower* endpoint a (so that division by $\ell \geq a$ is legitimate).

3 The lower bound 0.40

By Lemma 2.4 it suffices to exhibit one admissible law, one sample size and one threshold with normalized error at least $2/5$. The witness is a standardized Bernoulli variable with success probability $2/5$ and six summands. Every quantity in the argument is either an exact rational number, an exact multiple of $\sqrt{6}$, or a normal probability for which one rational upper bound is proved.

3.1 The witness law and its moments

Definition 3.1. Let B be Bernoulli with $\mathbb{P}(B = 1) = 2/5$ and set

$$X := \frac{5B - 2}{\sqrt{6}}, \quad \text{i.e.} \quad X = \frac{\sqrt{6}}{2} \text{ w.p. } \frac{2}{5}, \quad X = -\frac{\sqrt{6}}{3} \text{ w.p. } \frac{3}{5}.$$

Write μ^* for the law of X .

Lemma 3.2. μ^* is admissible, and

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \beta^* := \mathbb{E}|X|^3 = \frac{13\sqrt{6}}{30}, \quad \frac{\sqrt{6}}{\beta^*} = \frac{30}{13}.$$

Proof. Integrability is automatic for a two-point law. With $p = 2/5$, $q = 3/5$, $a = \sqrt{6}/2$, $b = -\sqrt{6}/3$ we have $pa + qb = \frac{2}{5} \cdot \frac{\sqrt{6}}{2} - \frac{3}{5} \cdot \frac{\sqrt{6}}{3} = \frac{\sqrt{6}}{5} - \frac{\sqrt{6}}{5} = 0$ and $pa^2 + qb^2 = \frac{2}{5} \cdot \frac{6}{4} + \frac{3}{5} \cdot \frac{6}{9} = \frac{3}{5} + \frac{2}{5} = 1$. For the third absolute moment, $p|a|^3 + q|b|^3 = \frac{2}{5} \cdot \frac{6\sqrt{6}}{8} + \frac{3}{5} \cdot \frac{6\sqrt{6}}{27} = \frac{3\sqrt{6}}{10} + \frac{2\sqrt{6}}{15} = \frac{9\sqrt{6} + 4\sqrt{6}}{30} = \frac{13\sqrt{6}}{30}$. Finally $\sqrt{6}/\beta^* = 30/13$. $\square$

LEAN. `witnessLaw`, `witness_admissible`, `witness_thirdAbsMoment`, `witness_normalizing_ratio` in `BerryEsseen/BernoulliWitness.lean`. The law is built as `mathlib's bernoulliMeasure` on the two atoms, so the product measure of Definition 2.2 is the six-fold product of the actual coin.

3.2 The event and its exact probability

Lemma 3.3. Let $X_1, \dots, X_6$ be independent copies of X and $W_6 = (X_1 + \dots + X_6)/\sqrt{6}$. If $K = B_1 + \dots + B_6$ denotes the number of successes, then

$$W_6 = \frac{5K - 12}{6}, \quad \text{so} \quad W_6 \leq -\frac{1}{3} \iff K \leq 2,$$

and

$$\mathbb{P}(W_6 \leq -\frac{1}{3}) = \sum_{k=0}^2 \binom{6}{k} \left(\frac{2}{5}\right)^k \left(\frac{3}{5}\right)^{6-k} = \frac{1701}{3125}.$$

Proof. $X_j = \frac{\sqrt{6}}{6}(5B_j - 2)$, so $\sum_j X_j = \frac{\sqrt{6}}{6}(5K - 12)$ and $W_6 = (5K - 12)/6$. Then $W_6 \leq -1/3$ iff $5K - 12 \leq -2$ iff $K \leq 2$. The probability is $3^6/5^6 + 6 \cdot 2 \cdot 3^5/5^6 + 15 \cdot 4 \cdot 3^4/5^6 = (729 + 2916 + 4860)/15625 = 8505/15625 = 1701/3125$. $\square$

LEAN. `normalizedSum_witness`, `normalizedSum_witness_le_iff`, `witness_sumLaw_eq_map`, `witness_six_probability` in `BerryEsseen/BernoulliWitness.lean`. Rather than invoking a binomial formula, the Lean proof evaluates the probability by summing over all $2^6 = 64$ Boolean outcomes of the product measure: `witness_finite_sum_rational` is a single exact rational identity

$$\sum_{b \in \{0,1\}^6} \mathbf{1}\{\#b \leq 2\} \prod_{i=1}^6 (b_i \frac{2}{5} + (1 - b_i) \frac{3}{5}) = \frac{1701}{3125}$$

discharged by `decide +kernel`. The passage from the pushforward of the product measure on $\mathbb{R}^6$ to the product measure on $\{0,1\}^6$ uses `mathlib's Measure.pi_map_pi` and `Measure.pi_singleton`.

3.3 One rational estimate of a normal probability

Lemma 3.4. $\Phi(-\frac{1}{3}) \leq \frac{1}{2} - \frac{1325}{10206}$.

Proof. By symmetry $\Phi(0) = \frac{1}{2}$, so $\Phi(-\frac{1}{3}) = \frac{1}{2} - \int_{-1/3}^0 \varphi$. On $[-\frac{1}{3}, 0]$ we have $e^{-x^2/2} \geq 1 - \frac{x^2}{2}$, and $\pi < 3.15$ gives $(2\pi)^{-1/2} \geq \frac{25}{63}$. Hence

$$\int_{-1/3}^0 \varphi \geq \frac{25}{63} \int_{-1/3}^0 \left(1 - \frac{x^2}{2}\right) dx = \frac{25}{63} \left(\frac{1}{3} - \frac{1}{162}\right) = \frac{25}{63} \cdot \frac{53}{162} = \frac{1325}{10206}. \quad \square$$

LEAN. `normal_density_coefficient_lower`, `normal_density_lower_on_small_interval`, `normalCDF_neg_third_upper` in `BerryEsseen/GaussianBounds.lean`; the bound $(2\pi)^{-1/2} \geq 25/63$ is derived from `mathlib`'s decimal bounds on π .

3.4 Proof of Theorem 1.1

Proof of Theorem 1.1. By Lemmas 3.2–3.4,

$$\frac{\sqrt{6}}{\beta^*} \left| \mathbb{P}(W_6 \leq -\frac{1}{3}) - \Phi(-\frac{1}{3}) \right| \geq \frac{30}{13} \left(\frac{1701}{3125} - \frac{1}{2} + \frac{1325}{10206} \right) = 0.401874\dots > \frac{2}{5}.$$

(The displayed rational number is $\frac{30}{13} \cdot \frac{1701 \cdot 2 \cdot 10206 - 3125 \cdot 10206 + 1325 \cdot 6250}{2 \cdot 3125 \cdot 10206}$, and the inequality with $2/5$ is a rational inequality.) Lemma 2.4 applied to μ^* , $n = 6$ and $x = -1/3$ gives $C_{\text{id}} \geq 2/5$. $\square$

LEAN. `lower_witness_normal_gap` in `BerryEsseen/GaussianBounds.lean` and `BerryEsseen.berryEsseenConstant_lower_bound` in `BerryEsseen/LowerBound.lean`. The Lean proof also carries out the conversion $(0.40 : \text{ENNRReal}) = \text{ENNRReal.ofReal}(2/5)$ explicitly.

Remark 3.5. The witness is chosen so that the lattice jump of W_6 straddles the threshold: W_6 takes the values $(5k - 12)/6$ for $k = 0, \dots, 6$, i.e. $-2, -\frac{7}{6}, -\frac{1}{3}, \frac{1}{2}, \frac{4}{3}, \frac{13}{6}, 3$, and $-1/3$ is an atom. Half of that atom's mass is the leading source of the discrepancy; the skewness of the asymmetric Bernoulli law contributes in the same direction. Section 3.5 makes both contributions quantitative in the limit $n \rightarrow \infty$.

3.5 Esseen's stronger lower bound (mathematics only, not formalized)

The repository also contains a complete mathematical proof of Esseen's lower bound [3]

$$C_{\text{id}} \geq c_E = \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = 0.4097321837\dots, \quad (2)$$

in `docs/ESSEEN_LOWER_BOUND.md`. **This asymptotic argument is not formalized in Lean**; the formalized lower bound is the weaker 0.40 of Theorem 1.1, and the upper-bound formalization does not use (2) in any way. We summarize the argument because it explains the numerical target and shows which features of the Bernoulli family produce it. The Bernoulli coefficient used below is the one recorded in the introduction and equation (7) of [18].

Fix a rational $p \in (0, \frac{1}{2})$, put $q = 1 - p$, $\sigma = \sqrt{pq}$, let B be Bernoulli(p) and $X = (B - p)/\sigma$. Then $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$ and

$$\beta(p) = \frac{pq^3 + qp^3}{(pq)^{3/2}} = \frac{p^2 + q^2}{\sqrt{pq}}.$$

Take n through multiples of the denominator of p , so that $m := np$ is an integer; at the threshold 0 we have $\mathbb{P}((X_1 + \dots + X_n)/\sqrt{n} \leq 0) = \mathbb{P}(K_n \leq m)$ with $K_n = \sum_j B_j$. The key asymptotic is

$$\mathbb{P}(K_n \leq np) = \frac{1}{2} + \frac{2 - p}{3\sqrt{2\pi npq}} + O_p(n^{-1}). \quad (3)$$

An exact integral for the binomial tail. For integers $0 \leq m < n$ let $G(t) = \sum_{j=0}^m \binom{n}{j} t^j (1-t)^{n-j}$. Differentiating and using $j \binom{n}{j} = n \binom{n-1}{j-1}$ and $(n-j) \binom{n}{j} = n \binom{n-1}{j}$, all terms cancel except the last:

$$G'(t) = -(n-m) \binom{n}{m} t^m (1-t)^{n-m-1}.$$

Since $G(0) = 1$ and $G(1) = 0$, integration gives the exact identity

$$\mathbb{P}(K_n \leq m) = G(p) = \frac{\int_p^1 t^m (1-t)^{n-m-1} dt}{\int_0^1 t^m (1-t)^{n-m-1} dt}. \quad (4)$$

No Stirling approximation is needed.

Laplace expansion at the common maximum. Let $L(t) = p \log t + q \log(1-t)$, so the integrand in (4) is $e^{nL(t)}/(1-t)$. Then L has its unique maximum at $t = p$, with

$$L'(p) = 0, \quad L''(p) = -\frac{1}{pq}, \quad L'''(p) = \frac{2(q-p)}{p^2q^2}.$$

Substituting $t = p + \sigma y/\sqrt{n}$, Taylor's theorem gives

$$n(L(p + \sigma y/\sqrt{n}) - L(p)) = -\frac{y^2}{2} + \frac{q-p}{3\sigma\sqrt{n}}y^3 + O_p(y^4/n), \quad \frac{q}{1-t} = 1 + \frac{\sigma}{q\sqrt{n}}y + O_p(y^2/n).$$

Writing $A_n(y) = \frac{q}{q-\sigma y/\sqrt{n}} \exp(n[L(p + \sigma y/\sqrt{n}) - L(p)])$ on $I_n = (-p\sqrt{n}/\sigma, q\sqrt{n}/\sigma)$, the common factor $e^{nL(p)}\sigma/(q\sqrt{n})$ cancels in the ratio (4) and

$$A_n(y) = e^{-y^2/2} \left(1 + \frac{P_p(y)}{\sqrt{n}}\right) + (\text{integrable error}), \quad P_p(y) = \frac{\sigma}{q}y + \frac{q-p}{3\sigma}y^3. \quad (5)$$

The polynomial P_p is odd: its contribution vanishes over the whole line but is positive over the positive half-line. That asymmetry is the entire source of the improvement over the symmetric example.

Why integrating (5) is legitimate. Pointwise Taylor expansion does not suffice; the error must be integrated, including the omitted tails. Choose $\delta > 0$ with $[p-\delta, p+\delta] \subset (0,1)$ small enough that $L(t) - L(p) \leq -c_p(t-p)^2$ on that interval and that $L^{(4)}$ and the first two derivatives of $q/(1-t)$ are bounded there. Split I_n at $R_n = n^{1/12}$ and at $\delta\sqrt{n}/\sigma$.

- *Central region* $|y| \leq R_n$. With $|e^z - 1 - z| \leq Cz^2$ for $|z| \leq 1$, the two expansions give

$$\left| A_n(y) - e^{-y^2/2} \left(1 + P_p(y)/\sqrt{n}\right) \right| \leq \frac{C_p}{n} e^{-y^2/2} (y^2 + y^4 + y^6),$$

where y^4/n comes from the cubic remainder, y^6/n from squaring the cubic term and y^2/n from the prefactor. The Gaussian moments on the right are integrable over all of $\mathbb{R}$, so the total contribution is $O_p(n^{-1})$.

- *Intermediate region* $R_n < |y| \leq \delta\sqrt{n}/\sigma$. Here $A_n(y) \leq C_p e^{-c_p\sigma^2 y^2}$, whose integral over $|y| > R_n$ is smaller than every inverse power of n because $R_n^2 = n^{1/6}$; the tails of the Gaussian polynomial in (5) have the same property.
- *Outer region, including the singular endpoint* $t \rightarrow 1$. Since L tends to $-\infty$ at 0 and 1 and has its unique maximum at p , there is $\eta_p > 0$ with $L(t) - L(p) \leq -\eta_p$ for $|t-p| \geq \delta$. Hence for $n \geq n_0$,

$$\int_{|t-p| \geq \delta} \frac{e^{n(L(t)-L(p))}}{1-t} dt \leq e^{-n\eta_p/2} \int_0^1 \frac{e^{(n_0/2)(L(t)-L(p))}}{1-t} dt,$$

and the last integral converges because near 0 the integrand behaves like $t^{n_0p/2}$ and near 1 like $(1-t)^{n_0q/2-1}$ with exponent > -1 . Changing variables multiplies the bound by $q\sqrt{n}/\sigma$, which leaves exponential decay.

Evaluating the coefficient. Since P_p is odd, $\int_{I_n} A_n = \sqrt{2\pi} + O_p(n^{-1})$. On the positive half-line $\int_0^\infty ye^{-y^2/2} dy = 1$ and $\int_0^\infty y^3 e^{-y^2/2} dy = 2$, so

$$\int_{I_n \cap [0, \infty)} A_n = \frac{\sqrt{2\pi}}{2} + \frac{1}{\sqrt{n}} \left(\frac{\sigma}{q} + \frac{2(q-p)}{3\sigma} \right) + O_p(n^{-1}) = \frac{\sqrt{2\pi}}{2} + \frac{2-p}{3\sigma\sqrt{n}} + O_p(n^{-1}),$$

using $\sigma/q = p/\sigma$. Dividing by the denominator of (4), which tends to $\sqrt{2\pi} > 0$, proves (3).

Optimizing. Since $\Phi(0) = \frac{1}{2}$, Lemma 2.4 and (3) give, for each fixed rational $p \in (0, \frac{1}{2})$,

$$C_{\text{iid}} \geq \mathcal{E}(p) := \frac{2-p}{3\sqrt{2\pi} [p^2 + (1-p)^2]}.$$

The function $\mathcal{E}$ is continuous and the rationals are dense, so the bound extends to all real $p \in (0, \frac{1}{2})$; in particular to $p_* = 2 - \frac{\sqrt{10}}{2}$. Note the order of limits: for each fixed rational p one first lets $n \rightarrow \infty$ through the allowed multiples, and only afterwards uses continuity of the explicit function $\mathcal{E}$; no nonuniform limits are interchanged and no integrality of np_* is needed. Differentiating $(2-p)/(2p^2-2p+1)$ gives a derivative with the sign of $2p^2 - 8p + 3$, positive before p_* and negative after it on $(0, \frac{1}{2})$, so p_* maximizes. At p_* one has $2 - p_* = \sqrt{10}/2$ and $p_*^2 + (1 - p_*)^2 = 10 - 3\sqrt{10}$, whence

$$C_{\text{iid}} \geq \mathcal{E}(p_*) = \frac{\sqrt{10}}{6\sqrt{2\pi}(10 - 3\sqrt{10})} = \frac{\sqrt{10} + 3}{6\sqrt{2\pi}} = c_E.$$

Where the improvement comes from. The coefficient in (3) can be rewritten

$$\frac{1}{\sqrt{2\pi pq}} \left(\frac{1}{2} + \frac{q-p}{6} \right) :$$

the term $\frac{1}{2}$ is the half-jump of the lattice distribution at its mean, and $(q-p)/6$ is the skewness correction. For $p < \frac{1}{2}$ the two reinforce each other at the inclusive threshold $K_n \leq np$. Dividing by $\beta(p)$ and balancing the two contributions produces p_* and the $\sqrt{10}$.

4 Cosine anchors: retaining the two-point law

This section and the next two develop the analytic estimates on which every region of the upper bound rests. Throughout, X satisfies (1), $Y = |X|$, $\delta = \beta - 1$ and $f(t) = \mathbb{E}e^{itX}$.

The point of the estimates is stated first. The classical third-order bound $|f(t) - (1 - t^2/2)| \leq \beta|t|^3/6$ is insensitive to the structure of the law. But $\beta = 1$ forces $|X| = 1$ a.s., and then the zero mean forces $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = \frac{1}{2}$ and $f(t) = \cos t$ exactly. An estimate that compares f with $\cos$ instead of with a quadratic therefore retains information that is lost otherwise, and every error term below carries a factor vanishing as $\delta \downarrow 0$.

4.1 The moment excess

Lemma 4.1. *Let $w(y) = (y-1)^2(2y+1) = 2y^3 - 3y^2 + 1$. Then $w \geq 0$ on $[0, \infty)$ and*

$$\boxed{\mathbb{E}w(Y) = 2\delta = 2(\beta - 1).}$$

Consequently $\beta \geq 1$ (so $\beta > 0$), with $\beta = 1$ if and only if $Y = 1$ almost surely. Moreover $\mathbb{E}Y \leq 1$.

Proof. Expanding, $\mathbb{E}w(Y) = 2\mathbb{E}Y^3 - 3\mathbb{E}Y^2 + 1 = 2\beta - 3 + 1 = 2(\beta - 1)$, using $\mathbb{E}Y^2 = \mathbb{E}X^2 = 1$ and $\mathbb{E}Y^3 = \beta$. Nonnegativity of w on $[0, \infty)$ is clear from the factored form. If $\delta = 0$ then $\mathbb{E}w(Y) = 0$ with $w(Y) \geq 0$, so $w(Y) = 0$ a.s., i.e. $Y = 1$ a.s. Finally $2Y \leq Y^2 + 1$ integrates to $2\mathbb{E}Y \leq 2$. $\square$

LEAN. `momentExcessWeight, integral_momentExcessWeight, one_le_thirdAbsMoment, thirdAbsMoment_pos, integral_abs_le_one, ae_abs_eq_one_of_thirdAbsMoment_eq_one` in `BerryEsseen/MomentExcess.lean`.

4.2 Two weighted Cauchy–Schwarz estimates

Lemma 4.2.

$$\boxed{\mathbb{E}[Y|Y^2 - 1|] \leq \sqrt{\delta(\beta + \frac{5}{3})}, \quad \mathbb{E}[Y|Y - 1|] \leq \sqrt{\delta}.$$

Proof. For the first, put $\rho(y) = \frac{y^3}{2} + \frac{7y^2}{9} + \frac{y}{18}$, so $\rho \geq 0$ on $[0, \infty)$. The polynomial identity

$$w(y)\rho(y) - (y|y^2 - 1|)^2 = \frac{y(y-1)^4}{18} \geq 0 \quad (y \geq 0)$$

holds pointwise. Writing $U = \sqrt{w(Y)}$, $V = \sqrt{\rho(Y)}$ and using $Y|Y^2 - 1| \leq UV$ pointwise, Cauchy–Schwarz gives

$$(\mathbb{E}[Y|Y^2 - 1|])^2 \leq (\mathbb{E}[UV])^2 \leq \mathbb{E}U^2 \cdot \mathbb{E}V^2 = \mathbb{E}w(Y) \cdot \mathbb{E}\rho(Y) = 2\delta \left(\frac{\beta}{2} + \frac{5}{6} \right) = \delta \left(\beta + \frac{5}{3} \right),$$

using $\mathbb{E}\rho(Y) = \frac{\beta}{2} + \frac{7}{9} + \frac{\mathbb{E}Y}{18} \leq \frac{\beta}{2} + \frac{5}{6}$ by $\mathbb{E}Y \leq 1$ and $\frac{7}{9} + \frac{1}{18} = \frac{15}{18} = \frac{5}{6}$.

For the second, the pointwise identity $w(y)\frac{y}{2} - (y|y-1|)^2 = \frac{y(y-1)^2}{2} \geq 0$ and Cauchy–Schwarz give $(\mathbb{E}[Y|Y-1|])^2 \leq \mathbb{E}w(Y) \cdot \frac{\mathbb{E}Y}{2} \leq \delta$. $\square$

LEAN. `cubicWeight`, `cubicWeight_identity`, `weighted_cubic_sq_le`, `integral_abs_weighted_sq_bound`, `weighted_linear_sq_le`, `integral_abs_weighted_linear_bound` in `BerryEsseen/WeightedMoments.lean`; the abstract Cauchy–Schwarz step in the form “a pointwise quadratic inequality gives an integrated one” is `integral_sq_le_mul_of_quadratic_nonneg` in `BerryEsseen/IntegralQuadratic.lean`. Only a third moment is used: all majorants have degree at most three, and the products are handled pointwise before integration.

4.3 One integral identity gives both sine estimates

Lemma 4.3. *For all $t, x \in \mathbb{R}$,*

$$\sin(tx) - x \sin t = tx \int_0^1 [\cos(txv) - \cos(tv)] dv. \quad (6)$$

Consequently

$$|\sin(tx) - x \sin t| \leq \frac{|t|^3}{6} |x| |x^2 - 1|, \quad (7)$$

$$|\sin(tx) - x \sin t| \leq \frac{t^2}{2} |x| ||x| - 1|. \quad (8)$$

Proof. Both sides of (6) vanish at $t = 0$ and have the same t -derivative, by the fundamental theorem of calculus applied to $v \mapsto \sin(txv)/(tx)$ and $v \mapsto \sin(tv)$. For (7) use $|\cos a - \cos b| \leq \frac{1}{2}|a^2 - b^2|$, which follows from the product-to-sum formula $\cos a - \cos b = -2 \sin \frac{a+b}{2} \sin \frac{a-b}{2}$ and $|\sin u| \leq |u|$; this makes the integrand at most $\frac{1}{2}t^2v^2|x^2 - 1|$, and $\int_0^1 v^2 dv = \frac{1}{3}$. For (8) use $|\cos a - \cos b| \leq |a - b|$, giving integrand at most $|t|v|x - 1|$ for $x \geq 0$ and $\int_0^1 v dv = \frac{1}{2}$; the case $x < 0$ follows by oddness of both sides in x combined with $|-x| = |x|$. $\square$

LEAN. `abs_cos_sub_cos_le_sq`, `sin_sub_mul_sin_eq_integral`, `abs_sin_sub_mul_sin_cubic`, `abs_sin_sub_mul_sin_quadratic` in `BerryEsseen/TrigonometricAnchors.lean`.

Proposition 4.4 (Imaginary anchor).

$$|\operatorname{Im} f(t)| \leq \min \left\{ \frac{|t|^3}{6} \sqrt{\delta(\beta + \frac{5}{3})}, \frac{t^2}{2} \sqrt{\delta}, \frac{\beta |t|^3}{6} \right\}. \quad (9)$$

Proof. $\operatorname{Im} f(t) = \mathbb{E} \sin(tX)$, and since $\mathbb{E}X = 0$ we may subtract $X \sin t$ without changing the expectation: $\operatorname{Im} f(t) = \mathbb{E}[\sin(tX) - X \sin t]$. Taking absolute values inside and applying (7), (8) and Lemma 4.2 gives the first two entries. The third is the classical $|\sin u - u| \leq |u|^3/6$ applied with $u = tX$, again using $\mathbb{E}X = 0$. $\square$

LEAN. `charFun_im_cubic_anchor`, `charFun_im_quadratic_anchor`, `charFun_im_classical_bound` in `BerryEsseen/ImaginaryPart.lean`; their minimum is packaged as `BerryEsseen.imaginaryBound` with `charFun_im_le_imaginaryBound` in `BerryEsseen/CharacteristicBounds.lean`.

4.4 Integrating the sine estimate gives the cosine anchor

Proposition 4.5 (Real anchor).

$$|\operatorname{Re} f(t) - \cos t| \leq \frac{\delta |t|^3}{6}. \quad (10)$$

Proof. For $y \geq 0$ define the remainder with a quadratic correction

$$R_t(y) := \cos(ty) - \cos t + \frac{t \sin t}{2} (y^2 - 1),$$

so that $R_t(1) = 0$ and $R'_t(y) = -t[\sin(ty) - y \sin t]$. By (8), $|R'_t(y)| \leq \frac{|t|^3}{2} y |y - 1|$. Integrating from 1 to y in the appropriate direction and using

$$\int_{\min(1,y)}^{\max(1,y)} s|s-1| ds = \frac{2y^3 - 3y^2 + 1}{6} = \frac{w(y)}{6}$$

gives $|R_t(y)| \leq \frac{|t|^3}{12} w(y)$ for all $y \geq 0$. Now substitute $y = Y = |X|$ and take expectations. Since $\cos(tY) = \cos(tX)$ and $\operatorname{Re} f(t) = \mathbb{E} \cos(tX)$, and since $\mathbb{E} Y^2 = 1$ kills the quadratic correction,

$$|\operatorname{Re} f(t) - \cos t| = |\mathbb{E} R_t(Y)| \leq \frac{|t|^3}{12} \mathbb{E} w(Y) = \frac{|t|^3}{12} \cdot 2\delta = \frac{\delta |t|^3}{6}. \quad \square$$

LEAN. `cosineRemainder`, `abs_cosineRemainder_le`, `integral_cosineRemainder`, `charFun_re_anchor` in `BerryEsseen/RealPart.lean`; the weight integral is `integral_linear_weight`. A pleasant consequence, proved as `charFun_eq_cos_of_thirdAbsMoment_eq_one` in `BerryEsseen/CharacteristicBounds.lean`, is that $\beta = 1$ implies $f(t) = \cos t$ for all t .

4.5 Combining the coordinates

Propositions 4.4 and 4.5 bound the two coordinates of $f(t)$ separately; combining them in the Euclidean norm gives bounds on $|f(t)|$ and on $|f(t) - g(t)|$ that are better near $\beta = 1$ than either coordinate alone would suggest.

Lemma 4.6. *If $|\operatorname{Re} z| \leq a$ and $|\operatorname{Im} z| \leq b$ then $|z| \leq \sqrt{a^2 + b^2}$. Consequently, writing $I(\beta, t)$ for the right-hand side of (9),*

$$|f(t)| \leq \min \left\{ 1, \sqrt{(|\cos t| + \frac{\delta |t|^3}{6})^2 + I(\beta, t)^2} \right\}, \quad (11)$$

$$|f(t) - c| \leq \sqrt{(|\cos t - c| + \frac{\delta |t|^3}{6})^2 + I(\beta, t)^2} \quad (c \in \mathbb{R}). \quad (12)$$

LEAN. `norm_le_sqrt_of_coordinate_bounds`, `norm_charFun_anchor`, `norm_charFun_sub_real_anchor` in `BerryEsseen/CharacteristicBounds.lean`.

Two comparison functions are used for c in (12): the Gaussian $c = g(t) = e^{-t^2/2}$, for which one needs the Gaussian–cosine discrepancy of Lemma 4.7 below, and, in the finite-sample region, nothing else.

Lemma 4.7 (Gaussian versus cosine). *For all real u ,*

$$e^{-u^2/2} - \cos u = \int_0^u \sin(u-v) v^2 e^{-v^2/2} dv,$$

and hence for $0 \leq u \leq \pi$,

$$0 \leq e^{-u^2/2} - \cos u \leq \frac{u^4}{12} \leq \frac{u^4}{8}.$$

There is also an explicit polynomial refinement $0 \leq e^{-u^2/2} - \cos u \leq u^4 Q(u^2)$ on $[0, \pi]$ with Q rational and nonnegative.

Proof. The function $F(v) = (\cos(u-v) - v \sin(u-v))e^{-v^2/2}$ has derivative $F'(v) = \sin(u-v) v^2 e^{-v^2/2}$, while $F(0) = \cos u$ and $F(u) = e^{-u^2/2}$; the identity is $\int_0^u F' = F(u) - F(0)$. For $0 \leq v \leq u \leq \pi$ we have $\sin(u-v) \geq 0$, whence nonnegativity. For the upper bound use $\sin(u-v) \leq u-v$ and $e^{-v^2/2} \leq 1$ and $\int_0^u (u-v) v^2 dv = u^4/12$. $\square$

LEAN. `gaussian_cosine_integral`, `gaussian_cosine_nonneg`, `gaussian_cosine_le_quartic` in `BerryEsseen/GaussianCosine.lean`; the polynomial refinement is `gaussianCosineQ` with `gaussian_cosine_le_Q` in `BerryEsseen/GaussianPolynomial.lean`.

Definition 4.8 (The two all-frequency majorants). For $\beta \geq 1$ and $t \in \mathbb{R}$ set

$$M(\beta, t) := \min \left\{ \min \left\{ 1, \sqrt{(|\cos t| + \frac{(\beta-1)|t|^3}{6})^2 + I(\beta, t)^2} \right\}, \sqrt{1 - 2t^2 h((\beta+1)|t|)} \right\},$$

$$E(\beta, t) := \min \left\{ \sqrt{(|\cos t - e^{-t^2/2}| + \frac{(\beta-1)|t|^3}{6})^2 + I(\beta, t)^2}, \frac{\beta |t|^3}{6} + \frac{t^4}{8} \right\},$$

where h is the convex minorant of Section 5. Then $|f(t)| \leq M(\beta, t)$ and $|f(t) - g(t)| \leq E(\beta, t)$ for every admissible law with third absolute moment β , at every real frequency.

LEAN. `BerryEsseen.modulusBound`, `BerryEsseen.errorBound`, `norm_charFun_le_modulusBound`, `norm_charFun_sub_gaussian_le_errorBound` in `BerryEsseen/SumBounds.lean`. The second entry of E is the classical estimate, proved in `BerryEsseen/ClassicalError.lean` as `norm_charFun_sub_gaussian_classical`. Each minimum combines independently valid inequalities and no cutoff restriction on t is imposed.

5 A global exponential modulus bound

The estimate of this section is what makes the argument valid at *all* frequencies, including for lattice laws whose characteristic functions return to modulus one. It is the reason no “Gaussian decay outside a small interval” assumption appears anywhere.

Theorem 5.1. *Let X satisfy (1) and let h be the function of Definition 5.3. Then for all $t \in \mathbb{R}$,*

$$|f(t)|^2 \leq 1 - 2t^2 h((\beta + 1)|t|), \quad |f(t)| \leq \exp\{-t^2 h((\beta + 1)|t|)\}.$$

LEAN. `norm_charFun_sq_minorant` and `norm_charFun_le_minorant_exp` in `BerryEsseen/FullModulus.lean`.

5.1 Symmetrization

Lemma 5.2. *Let X' be an independent copy of X and $D = X - X'$. Then*

$$|f(t)|^2 = \mathbb{E} \cos(tD), \quad \mathbb{E} D^2 = 2, \quad \mathbb{E} |D|^3 \leq 2(\beta + 1).$$

Proof. $|f(t)|^2 = f(t)\overline{f(t)} = \mathbb{E} e^{itX} \mathbb{E} e^{-itX'} = \mathbb{E} e^{itD} = \mathbb{E} \cos(tD)$, the imaginary part vanishing by symmetry of D . Also $\mathbb{E} D^2 = \mathbb{E} X^2 + \mathbb{E} X'^2 - 2(\mathbb{E} X)^2 = 2$. For the third moment, the pointwise inequality $|X - X'| \leq |X| + |X'|$ gives $|D|^3 \leq (X - X')^2(|X| + |X'|)$, and expanding

$$\mathbb{E}[(X - X')^2(|X| + |X'|)] = 2\mathbb{E}|X|^3 + 2\mathbb{E}|X| = 2\beta + 2\mathbb{E}|X| \leq 2(\beta + 1),$$

where the cross terms containing $\mathbb{E} X$ or $\mathbb{E} X'$ vanish, the remaining ones use $\mathbb{E} X^2 = 1$, and $\mathbb{E}|X| \leq 1$ by Cauchy–Schwarz (or by Lemma 4.1). $\square$

LEAN. `integral_cos_symm`, `integral_symm_sq`, `integral_symm_abs_cube_le` in `BerryEsseen/Symmetrization.lean`; the integrability of the symmetrized cube uses the elementary majorant $|x - y|^3 \leq 4(|x|^3 + |y|^3)$.

Symmetrization converts a complex modulus into a real cosine expectation while retaining second- and third-moment information — exactly the two pieces of data the minorant argument consumes.

5.2 The convex minorant

Definition 5.3. Let $r = \frac{9}{2}$, $G(s) = \frac{1 - \cos s}{s^2}$ for $s > 0$, and

$$a := \frac{81}{80(1 - \cos r)}, \quad h(s) := \begin{cases} \frac{1}{2} - \frac{s}{10}, & s \leq r, \\ aG(s), & r < s \leq 2\pi, \\ 0, & s > 2\pi. \end{cases}$$

The linear branch is used also for $s < 0$, so that h is defined and convex on all of $\mathbb{R}$; only nonnegative arguments enter the probabilistic estimate.

Proposition 5.4. *The function h is convex on $\mathbb{R}$, nonnegative, nonincreasing, $h(s) \leq \frac{1}{2}$ for $s \geq 0$, $h(s) \leq G(s)$ for $s > 0$, and $\frac{5}{6} < a \leq 1$. Consequently, for all real v ,*

$$\cos v \leq 1 - v^2 h(|v|). \tag{13}$$

Proof sketch. The pieces are checked as follows.

1. *Linear branch below G .* The global inequality $\cos s - 1 + \frac{s^2}{2} \leq \frac{|s|^3}{10}$ rearranges to $G(s) \geq \frac{1}{2} - \frac{s}{10}$ for $s > 0$.
2. *Scale.* Elementary sine bounds at $\frac{3\pi}{2} - r$ give $-\frac{3}{14} \leq \cos r \leq -\frac{1}{8}$, so $\frac{5}{6} < a \leq 1$; hence $aG \leq G$, and the zero branch is trivially below G .
3. *Continuity at the joins.* $aG(r) = \frac{1}{20} = \frac{1}{2} - \frac{r}{10}$ and $G(2\pi) = 0$.

4. *Monotonicity and convexity of G on $[r, 2\pi]$.* Differentiating,

$$G'(s) = \frac{s \sin s - 2(1 - \cos s)}{s^3} \leq 0, \quad G''(s) = \frac{(s^2 - 6) \cos s - 4s \sin s + 6}{s^4} \geq 0.$$

For G'' , split at $\frac{3\pi}{2}$: on $[r, \frac{3\pi}{2}]$ the distance to $\frac{3\pi}{2}$ is at most $\frac{1}{2}$, so $\sin s \leq -\frac{3}{4}$ by $\cos v \geq 1 - \frac{v^2}{2}$, and with $r \leq s \leq 5$, $\cos s \geq -1$, the numerator is at least $12 - s^2 + 3s \geq \frac{1}{2}$; on $[\frac{3\pi}{2}, 2\pi]$ every nonconstant term of the numerator is nonnegative.

5. *No downward jump of slope.* At the first join $G'(r) \geq -2G(r)$, hence $aG'(r) \geq -\frac{1}{10}$, the slope of the linear branch; at the second join $G'(2\pi) = 0$. The Lean proof does not assume differentiability at a join: it constructs an explicit supporting line at every point and applies the criterion “a function with a supporting line everywhere is convex”.

Finally (13) is exactly $h(|v|) \leq G(|v|)$ for $v \neq 0$, and trivial at $v = 0$. $\square$

LEAN. `cosineProfile` and its two derivative lemmas in `BerryEsseen/CosineProfile.lean`; the splicing criterion `convexOn_of_supporting_lines` and `convexOn_convexSplice` in `BerryEsseen/ConvexSplice.lean`; `minorantScale`, `cosineMinorant`, `cos_splice_bounds`, `minorantScale_bounds`, `convexOn_cosineMinorant`, `antitone_cosineMinorant`, `cosineMinorant_le_half`, `cosineMinorant_supporting_line`, `cos_le_minorant` in `BerryEsseen/FullMinorant.lean`; the cubic cosine inequality `cosine_tenth` in `BerryEsseen/CosineMinorant.lean`.

5.3 One weighted Jensen step

Proof of Theorem 5.1. Put $z = (\beta + 1)|t|$ and choose, by convexity, a supporting line to h at z : $h(y) \geq h(z) + m(y - z)$ for all y , with $m \leq 0$ because h is nonincreasing. By Lemma 5.2 and (13) applied with $v = tD$,

$$1 - |f(t)|^2 = \mathbb{E}[1 - \cos(tD)] \geq t^2 \mathbb{E}[D^2 h(|t| |D|)].$$

Using the supporting line with $y = |t| |D|$,

$$\mathbb{E}[D^2 h(|t| |D|)] \geq h(z) \mathbb{E}D^2 + m(|t| \mathbb{E}|D|^3 - z \mathbb{E}D^2) = 2h(z) + m(|t| \mathbb{E}|D|^3 - 2z) \geq 2h(z),$$

since $|t| \mathbb{E}|D|^3 \leq 2(\beta + 1)|t| = 2z$ and $m \leq 0$. This is the first assertion. All integrals exist: h is bounded on $[0, \infty)$ and the supporting-line expression uses only the second and third moments already established. The second assertion follows from $1 - v \leq e^{-v}$ and taking square roots. $\square$

Remark 5.5. The step above is Jensen’s inequality with weights $D^2/2$, written so that no new probability measure has to be constructed. It is also where the specific combination $(\beta + 1)|t|$ arises: it is precisely $\frac{1}{2}|t| \mathbb{E}|D|^3 / \mathbb{E}D^2$ up to the bound of Lemma 5.2.

Since h vanishes for $s > 2\pi$, Theorem 5.1 gives no decay for $(\beta + 1)|t| > 2\pi$. This is unavoidable — for the symmetric two-point law $|f(2\pi k)| = 1$ — and it is the reason the smoothing cutoff T is always chosen so that the largest frequency $T\tau/\sqrt{n}$ entering the bound stays inside the useful range.

6 Prawitz smoothing, proved from scratch

Every region of the upper bound uses the same smoothing inequality: for any cutoff $T > 0$ and any split $0 < \tau \leq 1$, the Kolmogorov distance between a law with a finite first moment and the standard normal law is bounded by four explicit positive integrals in which the law appears only through its characteristic function on the bounded band $[0, T]$. This is Prawitz’s inequality [6]; the development proves it, including all integrability and endpoint issues, without importing it.

6.1 A pointwise majorant of the step function

Definition 6.1. For $0 < s < 1$ let $B(s) = (1 - s) \cot(\pi s) + \frac{1}{\pi}$ as in Convention 1.3 and set

$$H(x) := 2 \int_0^1 B(s) \sin(2\pi xs) ds, \quad S(x) := \frac{1 + H(x) + \operatorname{sinc}(\pi x)^2}{2}.$$

Although B has a pole at $s = 0$, the product $B(s) \sin(2\pi xs)$ is integrable: Jordan’s inequality on the two half-intervals $(0, \frac{1}{2}]$ and $[\frac{1}{2}, 1)$ gives

$$|B(s) \sin(2\pi xs)| \leq \pi|x| + 2 \quad (0 < s < 1), \quad (14)$$

so H is well defined, odd, and $H(0) = 0$; endpoint values of the integrand are irrelevant to the integral.

Theorem 6.2. For all $x \in \mathbb{R}$,

$$\boxed{\mathbf{1}_{[0,\infty)}(x) \leq S(x).}$$

Moreover $S(0) = 1$, S is measurable, and $|S(x)| \leq \pi|x| + 3$.

The proof has three steps, each elementary.

Step 1: a shift recurrence. The sine difference identity cancels the cotangent denominator:

$$\cot(\pi s) [\sin(2\pi(x+1)s) - \sin(2\pi xs)] = \cos(2\pi(x+1)s) + \cos(2\pi xs).$$

Only two elementary integrals then remain,

$$\int_0^1 \sin(ks) ds = \frac{1 - \cos k}{k}, \quad \int_0^1 (1-s) \cos(ks) ds = \frac{1 - \cos k}{k^2} \quad (k \neq 0),$$

and substitution and cancellation give, for $x > 0$,

$$\boxed{H(x+1) - H(x) = \frac{c(x)}{x^2(x+1)^2}, \quad c(x) := \left(\frac{\sin \pi x}{\pi}\right)^2 \geq 0.} \quad (15)$$

The same identity at $x = 0$ gives $H(1) = 1$, and since $c(n) = 0$ for every integer, $H(n) = 1$ for every positive integer n .

LEAN. `integral_sin_linear`, `integral_triangular_cos`, `prawitz_kernel_shift_identity`, `prawitz_integrand_bound`, `prawitzSineApprox_shift` in `BerryEsseen/PrawitzKernel.lean`; `prawitzSineApprox_one`, `prawitzSineApprox_nat_succ` in `BerryEsseen/PrawitzMajorant.lean`. The Lean name for H is `prawitzSineApprox` and for B is `prawitzSineKernel`.

Step 2: the limit needs only Riemann–Lebesgue. Fix $x \in \mathbb{R}$. Iterating the difference formula, or rather applying the integrand identity directly,

$$H(x+n) - H(n) = 4 \int_0^1 B(s) \sin(\pi xs) \cos(2\pi s(n+x/2)) ds.$$

The amplitude $s \mapsto B(s) \sin(\pi xs)$ is integrable on $(0,1)$ by (14), so the Riemann–Lebesgue lemma gives $H(x+n) - H(n) \rightarrow 0$; since $H(n) = 1$,

$$H(x+n) \rightarrow 1 \quad (n \rightarrow \infty).$$

No improper sine integral is evaluated and no cotangent Fourier series is summed.

LEAN. `tendsto_integral_mul_cos`, `tendsto_interval_mul_cos` in `BerryEsseen/OscillatoryIntegral.lean` (the real interval form of mathlib’s Riemann–Lebesgue theorem); `tendsto_prawitzSineApprox_nat_shift` in `BerryEsseen/PrawitzMajorant.lean`.

Step 3: two monotone sequences. Fix $x > 0$ and write $c = c(x) \geq 0$; integer shifts leave c unchanged. By (15) the sequence $n \mapsto H(x+n)$ is nondecreasing with limit 1, hence

$$H(x) \leq 1.$$

Adding the correction $c/(x+n)^2$ produces a nonincreasing sequence: a direct computation from (15) gives

$$\left[H(x+n+1) + \frac{c}{(x+n+1)^2} \right] - \left[H(x+n) + \frac{c}{(x+n)^2} \right] = -\frac{2c}{(x+n)(x+n+1)^2} \leq 0.$$

This corrected sequence dominates $H(x+n)$ and has the same limit 1, so

$$1 \leq H(x) + \frac{c(x)}{x^2} = H(x) + \text{sinc}(\pi x)^2,$$

that is, $S(x) \geq 1$ for $x > 0$. For $x < 0$ oddness of H together with $H(-x) \leq 1$ gives $H(x) \geq -1$, hence $S(x) \geq 0$; and $S(0) = 1$ because $H(0) = 0$ and $\text{sinc}(0) = 1$. This proves Theorem 6.2; the growth bound $|S| \leq \pi|x| + 3$ follows from (14) and $|\text{sinc}| \leq 1$.

LEAN. The abstract two-sequence argument is `bounds_of_kernel_recurrence` in `BerryEsseen/KernelRecurrence.lean`; `prawitzStepApprox`, `step_le_prawitzStepApprox`, `abs_prawitzStepApprox_le`, `measurable_prawitzStepApprox` in `BerryEsseen/PrawitzMajorant.lean`.

6.2 From the majorant to two-sided bounds on a distribution function

Proposition 6.3. *Let ν be a probability measure on $\mathbb{R}$ with $\int |y| d\nu < \infty$, let $F(a) = \nu((-\infty, a])$ and let $k > 0$. Then*

$$1 - \int S(k(y - a)) d\nu(y) \leq F(a) \leq \int S(k(a - y)) d\nu(y).$$

Proof. Apply Theorem 6.2 pointwise at $x = k(a - y)$: the left side is $\mathbf{1}\{y \leq a\}$, and integrating gives the upper bound. The lower bound is the reflected inequality. Both expectations exist by the growth bound $|S(x)| \leq \pi|x| + 3$ and the assumed first moment. An atom at a is handled explicitly: the upper pointwise bound equals 1 at $x = 0$ and the reflected lower bound equals 0 there, which is exactly what right-continuity of F requires. $\square$

LEAN. `cdf_le_integral_prawitzStepApprox,` `integral_prawitzStepApprox_reflected_le_cdf` in
BerryEsseen/PrawitzProbability.lean.

6.3 The Fourier form

Interchanging the probability integral with the s -integral in Definition 6.1 turns Proposition 6.3 into a statement about the characteristic function. The interchange is justified by the bound (14), which supplies an integrable majorant linear in $|y|$; in addition

$$\text{sinc}(\pi z)^2 = 2 \int_0^1 (1 - s) \cos(2\pi z s) ds.$$

Proposition 6.4. *Let ν be a probability measure with finite first moment and characteristic function $\hat{\nu}$, and let $T > 0$. With K_σ as in Convention 1.3,*

$$\frac{1}{2} + \int_0^1 \text{Re}\{K_{-1}(s)e^{-iTsx}\hat{\nu}(Ts)\}ds \leq F(x) \leq \frac{1}{2} + \int_0^1 \text{Re}\{K_{+1}(s)e^{-iTsx}\hat{\nu}(Ts)\}ds.$$

Proof sketch. Take $k = T/(2\pi)$ in Proposition 6.3, expand $S(k(x - y))$ using Definition 6.1 and the sinc^2 representation, and evaluate the resulting sine and cosine expectations as real parts of $\hat{\nu}$. The two kernels differ only in the sign of the real part, which is exactly the effect of reflecting $x - y \mapsto y - x$. $\square$

LEAN. `prawitzIntegrand,` `integral_prawitzIntegrand_prod,` `integral_integral_prawitzIntegrand_swap,`
`cdf_le_prawitz_doubleIntegral,` `prawitz_doubleIntegral_le_cdf` in BerryEsseen/PrawitzFubini.lean; `prawitzKernel,`
`cdf_le_prawitzFourierIntegral,` `prawitzFourierIntegral_le_cdf` in BerryEsseen/PrawitzFourier.lean.

6.4 The normal distribution function as one Fourier integral

Lemma 6.5. *For all $x \in \mathbb{R}$,*

$$\Phi(x) - \frac{1}{2} = \frac{1}{\pi} \int_0^\infty e^{-t^2/2} \frac{\sin(xt)}{t} dt = \frac{T}{\pi} \int_0^\infty e^{-T^2s^2/2} \frac{\sin(xTs)}{Ts} ds \quad (T > 0).$$

Proof. Let γ be the standard Gaussian measure. Fubini's theorem on the finite rectangle $[0, x] \times \mathbb{R}$, legitimate because $|\cos(ut)| \leq 1$, gives

$$\int_{\mathbb{R}} \frac{\sin(xt)}{t} d\gamma(t) = \int_0^x \left(\int_{\mathbb{R}} \cos(ut) d\gamma(t) \right) du = \int_0^x e^{-u^2/2} du,$$

the frequency $t = 0$ being a γ -null set. Substituting the Gaussian density and using evenness gives the first identity; the integral converges absolutely since $|\sin(xt)/t| \leq |x|$. The substitution $t = Ts$ gives the second. $\square$

LEAN. `integral_cos_standardGaussian,` `integral_sine_quotient_standardGaussian,` `normalCDF_fourier_full,`
`normalCDF_fourier_positive,` `normalCDF_fourier_scaled` in BerryEsseen/GaussianInversion.lean.

6.5 The four-term smoothing inequality

Theorem 6.6 (Prawitz smoothing). *Let ν be a probability measure on $\mathbb{R}$ with finite first moment and characteristic function $\hat{\nu}$, let $T > 0$ and $0 < \tau \leq 1$, and write $g_T(s) = e^{-T^2 s^2/2}$. Then for every $x \in \mathbb{R}$,*

$$\begin{aligned} |F(x) - \Phi(x)| &\leq \int_0^\tau |K(s)| |\hat{\nu}(Ts) - g_T(s)| ds && \text{(low-frequency error)} \\ &+ \int_\tau^1 |K(s)| |\hat{\nu}(Ts)| ds && \text{(high-frequency magnitude)} \\ &+ \int_0^\tau \left| K(s) - \frac{i}{\pi s} \right| g_T(s) ds && \text{(normal correction)} \\ &+ \frac{1}{\pi} \int_\tau^\infty \frac{g_T(s)}{s} ds && \text{(Gaussian tail).} \end{aligned}$$

Proof sketch. Subtract Lemma 6.5 from Proposition 6.4. On $(0, \tau)$ use the algebraic identity

$$K(s)\hat{\nu}(Ts) - \frac{i}{\pi s}g_T(s) = K(s)(\hat{\nu}(Ts) - g_T(s)) + \left(K(s) - \frac{i}{\pi s}\right)g_T(s),$$

which separates the distributional error from the correction to the normal inversion kernel; multiplication by e^{-iTsx} preserves norms, and a real part is bounded by a modulus. On $(\tau, 1)$ bound the distributional term by $|K(s)| |\hat{\nu}(Ts)|$. The remaining normal integral over $[\tau, \infty)$ is bounded using $|\sin(Tsx)| \leq 1$. The reflected kernel gives the same two norms because $K_{-1} = -\overline{K_{+1}}$ and $K_{-1} - \frac{i}{\pi s} = -\overline{K_{+1} - \frac{i}{\pi s}}$, so both inequalities of Proposition 6.4 yield the same bound.

The apparent singularity at $s = 0$ is removable. Jordan's inequality and elementary Taylor bounds give $|\cot(\pi s) - \frac{1}{\pi s}| \leq \frac{\pi^2 s}{3}$ for $0 < s \leq \frac{1}{2}$, so $K(s) - \frac{i}{\pi s}$ stays bounded and $s|K(s)|$ stays bounded; and a finite first moment gives

$$|\hat{\nu}(Ts) - g_T(s)| \leq \left(|T| \int |y| d\nu + \frac{T^2}{2}\right)s \quad (0 \leq s \leq 1),$$

which cancels the $1/s$ of the kernel. Away from zero the kernel norms are bounded and the last integral is dominated by a Gaussian divided by $\pi\tau$. $\square$

LEAN. `BerryEsseen.smoothingIntegral` and `BerryEsseen.prawitz_smoothing` in `BerryEsseen/PrawitzSmoothing.lean`; the four integrands are `prawitzLowError`, `prawitzHighMagnitude`, `prawitzNormalCorrection`, `prawitzNormalTail` in `BerryEsseen/SmoothingIntegrability.lean`, and the splitting lemma is in `BerryEsseen/SmoothingSplit.lean`.

For use with explicit majorants there is a corollary that substitutes any integrable upper bounds and requires them only on *open* intervals, so that endpoint values never matter.

Corollary 6.7. *In the situation of Theorem 6.6, let $R, M : \mathbb{R} \rightarrow \mathbb{R}$ satisfy $|\hat{\nu}(Ts) - g_T(s)| \leq R(s)$ for $s \in (0, \tau)$ and $|\hat{\nu}(Ts)| \leq M(s)$ for $s \in (\tau, 1)$, with $s \mapsto |K(s)|R(s)$ integrable on $(0, \tau)$ and $s \mapsto |K(s)|M(s)$ integrable on $(\tau, 1)$. Then $|F(x) - \Phi(x)|$ is at most the same four-term expression with $|\hat{\nu}(Ts) - g_T(s)|$ and $|\hat{\nu}(Ts)|$ replaced by $R(s)$ and $M(s)$.*

LEAN. `prawitz_smoothing_of_majorants` and, specialized to the normalized i.i.d. sum, `normalizedSum_prawitz_smoothing` in `BerryEsseen/SmoothingMajorants.lean`; the required first moment of μ_n comes from `integrable_id_normalizedSumLaw`.

6.6 Exact kernel norms and the function q

The two kernel norms appearing in Theorem 6.6 have closed forms in terms of $q(x) = \frac{1}{x} - \cot x$.

Lemma 6.8. *For $0 < s < 1$,*

$$\left| K(s) - \frac{i}{\pi s} \right| = (1-s)\sqrt{1+q(\pi s)^2}, \quad |K(s)| = (1-s)\sqrt{1+q(\pi(1-s))^2}.$$

Moreover q is nonnegative and increasing on $(0, \pi)$, so $|K|$ is decreasing on $(0, 1)$.

Proof. $\operatorname{Im} K(s) = (1-s)\cot(\pi s) + \frac{1}{\pi}$. Using $\cot(\pi(1-s)) = -\cot(\pi s)$,

$$(1-s)q(\pi(1-s)) = \frac{1}{\pi} + (1-s)\cot(\pi s) = \operatorname{Im} K(s),$$

which with $\operatorname{Re} K(s) = 1 - s$ gives the second identity. For the first, $\operatorname{Im}(K(s) - \frac{i}{\pi s}) = (1 - s) \cot(\pi s) + \frac{1}{\pi} - \frac{1}{\pi s} = -(1 - s)q(\pi s)$, as one checks by expanding $q(\pi s) = \frac{1}{\pi s} - \cot(\pi s)$. Nonnegativity and monotonicity of q follow from

$$q'(x) = \frac{1}{\sin^2 x} - \frac{1}{x^2} \geq 0, \quad \sin x - x \cos x = \int_0^x t \sin t \, dt \geq 0. \quad \square$$

Rational two-sided bounds for q are also needed, both for the analytic regions and as the meaning of the certificate tables. They come from signed Taylor bounds for sine, integrated against t .

Lemma 6.9. *Put*

$$\begin{aligned} A_-(x) &= \frac{1}{3} - \frac{x^2}{30} + \frac{x^4}{840} - \frac{x^6}{45360}, & A_+(x) &= A_-(x) + \frac{x^8}{3991680}, \\ D_+(x) &= 1 - \frac{x^2}{6} + \frac{x^4}{120} - \frac{x^6}{5040} + \frac{x^8}{362880}, & D_-(x) &= D_+(x) - \frac{x^{10}}{39916800}. \end{aligned}$$

Then $x^3 A_-(x) \leq \sin x - x \cos x \leq x^3 A_+(x)$ and $x D_-(x) \leq \sin x \leq x D_+(x)$, and consequently

$$q_-(x) := \frac{x A_-(x)}{D_+(x)} \leq q(x) \leq \frac{x A_+(x)}{D_-(x)} =: q_+(x), \quad (16)$$

the lower comparison requiring $A_- \geq 0$ and the upper one $D_- > 0$. Both denominators and the lower numerator are strictly positive on $[0, \frac{\pi}{2}]$.

LEAN. The four polynomials and the real inequalities are in `BerryEsseen/CotangentPolynomials.lean`; the positivity of the denominators on the closed half-period is in `BerryEsseen/FiniteKernels.lean`; the passage from the saved integer tables to (16), including the conversion from rational bounds on π to the table argument, is `qEntry_lower_sound` and `qEntry_upper_sound` in `BerryEsseen/Numerics/CotangentTable.lean`; the panel inequalities derived from monotonicity of q and of $|K|$ are in `BerryEsseen/CotangentCorrection.lean`.

A coarse bound that is convenient in the small-fraction region is: for $0 < x < \pi$ with $1 - x^2/6 > 0$,

$$0 \leq q(x) \leq \frac{x}{3(1 - x^2/6)}, \quad (17)$$

which follows from $\sin x - x \cos x = \int_0^x t \sin t \, dt \leq x^3/3$ and $\sin x \geq x(1 - x^2/6)$.

Finally, the following reflected estimate is used for the high-frequency endpoint panels of Section 11.

Lemma 6.10. *For $0 < s < 1$,*

$$q(\pi(1 - s)) \leq \frac{1 + s}{\pi s}, \quad (18)$$

and the right-hand side is decreasing in $s > 0$.

Proof. Jordan's inequality $\sin t \geq 2t/\pi$ on $[0, \frac{\pi}{2}]$ gives $\sin x - x \cos x = \int_0^x t \sin t \, dt \geq \frac{2x^3}{3\pi}$; dividing by $x \sin x \leq x^2$ yields $q(x) \geq \frac{2x}{3\pi}$ on that half-period. For $s \leq \frac{1}{2}$ this gives $q(\pi s) \geq \frac{2s}{3} \geq \frac{s}{\pi(1-s)}$ (using $\pi \geq 3$), and substituting into the exact reflection identity

$$q(\pi(1 - s)) = \frac{1}{\pi s} + \frac{1}{\pi} + \frac{s}{\pi(1 - s)} - q(\pi s)$$

gives (18). For $s \geq \frac{1}{2}$, monotonicity of q gives $q(\pi(1 - s)) \leq q(\pi/2) = 2/\pi \leq (1 + s)/(\pi s)$. $\square$

LEAN. `BerryEsseen/FiniteDarboux.lean`.

7 Power comparison and parameter compression

Independence gives

$$\hat{\mu}_n(t) = f(t/\sqrt{n})^n,$$

so the low-frequency term of Theorem 6.6 requires an estimate of $|z^n - w^n|$ where $z = f(t/\sqrt{n})$ and $w = e^{-t^2/(2n)}$. This section explains how a single inequality is made valid for infinitely many n at once, which is what allows 467 cells to cover all $n \geq 20$.

LEAN. `charFun_normalizedSumLaw` in `BerryEsseen/IndependentSums.lean` proves $\hat{\mu}_n(t) = f(t/\sqrt{n})^n$ for the product-measure definition of μ_n , and `gaussian_normalized_power` in `BerryEsseen/SumBounds.lean` proves $(e^{-(t/\sqrt{n})^2/2})^n = e^{-t^2/2}$.

7.1 Keeping the damping in the geometric sum

The elementary factorization is

$$|z^n - w^n| \leq |z - w| \sum_{j=0}^{n-1} |z|^{n-1-j} |w|^j. \quad (19)$$

The two bases decay at different rates, and bounding every term of the sum by its largest value would throw away the extra Gaussian decay of w . The following device retains it.

Definition 7.1. Put

$$A(x) := \int_0^1 e^{-xs} ds, \quad G_q(x) := \frac{A(x)}{A(qx)}.$$

A is positive, decreasing, continuous, and for $x \neq 0$ equals $(1 - e^{-x})/x$; hence for $q, x \neq 0$

$$G_q(x) = \frac{q(1 - e^{-x})}{1 - e^{-qx}}, \quad G_q(0) = 1.$$

The definition through A avoids the removable singularity at $x = 0$, which matters because the certificates evaluate G at rational arguments that may be zero.

Lemma 7.2. For $x \geq 0$: $q \mapsto G_q(x)$ is nondecreasing; $0 < G_q(x) \leq 1$ when $0 < q \leq 1$; and for fixed $0 < q \leq 1$, $x \mapsto G_q(x)$ is nonincreasing on $[0, \infty)$.

Proof. The first two assertions are immediate from positivity and monotonicity of A . For the third, differentiate the quotient formula: the sign of the derivative is the sign of

$$e^{-x}(1 - e^{-qx}) - qe^{-qx}(1 - e^{-x}),$$

and multiplying by $e^{(1+q)x} > 0$ reduces this to $e^{qx} - 1 - q(e^x - 1) \leq 0$, which is exactly the convexity inequality $e^{qx} \leq 1 - q + qe^x$ for $0 \leq q \leq 1$. Continuity extends the conclusion to $x = 0$. $\square$

LEAN. expAverage, geometricDamping, expAverage_formula, geometricDamping_le_one, geometricDamping_mono_q, geometricDamping_deriv_nonpos, antitoneOn_geometricDamping, geometricDamping_geometric_sum in BerryEsseen/GeometricDamping.lean.

Proposition 7.3. Let $\gamma = 1/n$, let $d > 0$ satisfy $(\beta + 1)/\sqrt{n} \leq d$, and put

$$\Psi(t) := t^2 h(d|t|), \quad D(t) := \frac{t^2}{2} - \Psi(t) \geq 0.$$

Then $|z| \leq e^{-\gamma\Psi(t)}$ and $w = e^{-\gamma(\Psi(t)+D(t))}$, so $|z^n| \leq e^{-\Psi(t)}$ at every frequency. If moreover $|z - w| \leq \gamma E(t)$, then

$$|z^n - w^n| \leq E(t) e^{-(1-\gamma)\Psi(t)} G_\gamma(D(t)), \quad (20)$$

and consequently, for any cap $\gamma \leq v \leq 1$,

$$\boxed{|z^n - w^n| \leq E(t) e^{-(1-v)\Psi(t)} G_v(D(t))}. \quad (21)$$

Proof. The modulus bound is Theorem 5.1 rescaled, using that h is nonincreasing and $(\beta+1)|t|/\sqrt{n} \leq d|t|$; the nonnegativity $D \geq 0$ is $h \leq \frac{1}{2}$. Substituting the two modulus bounds into (19),

$$\sum_{j=0}^{n-1} |z|^{n-1-j} w^j \leq e^{-(1-\gamma)\Psi(t)} \sum_{j=0}^{n-1} e^{-j\gamma D(t)} = n e^{-(1-\gamma)\Psi(t)} G_\gamma(D(t)),$$

the last equality being the exact evaluation of the geometric sum in terms of A . Combining with $|z - w| \leq \gamma E(t) = E(t)/n$ gives (20). Since $\Psi \geq 0$, replacing γ by the larger v increases both $e^{-(1-\gamma)\Psi}$ and, by Lemma 7.2, $G_\gamma(D)$; this gives (21). $\square$

LEAN. dampingExponent, norm_charFun_scaled_le_damping, norm_charFun_normalized_le_damping, norm_charFun_normalized_sub_gaussian_le_scalar_cap in BerryEsseen/ScalarCompression.lean; norm_pow_sub_pow_le_damped, norm_pow_sub_pow_le_damped_cap in BerryEsseen/DampedPower.lean.

For numerical evaluation one further needs an upper bound for G_v at a rational argument. Integrating the elementary Taylor bounds for the exponential gives, for $x \geq 0$, $q \leq 1$ and $qx < 2$,

$$G_q(x) \leq \min\left\{1, \frac{1 - \frac{x}{2} + \frac{x^2}{6} - \frac{x^3}{24} + \frac{x^4}{120}}{1 - \frac{qx}{2}}\right\}, \quad (22)$$

and $G_q(x) \leq 1$ whenever $qx \geq 2$.

LEAN. `expAverage_lower`, `expAverage_upper`, `geometricDamping_le_polynomial`, `geometricDamping_le_min` in `BerryEsseen/GeometricDamping.lean`; the integer guard that implies (22) for a saved certificate is `geometricGuard` with `geometricGuard_sound` in `BerryEsseen/Numerics/GeometricWitness.lean`.

7.2 Two error prefactors

Two choices of E in Proposition 7.3 are used. Write b for an upper cap on $\ell = \beta/\sqrt{n}$ and v for an upper cap on $1/n$.

Classical prefactor. With $u = t/\sqrt{n}$, Definition 4.8 gives

$$n|f(u) - e^{-u^2/2}| \leq \frac{\ell|t|^3}{6} + \frac{t^4}{8n} \leq E_c(t) := \frac{b|t|^3}{6} + \frac{vt^4}{8}.$$

Vector prefactor. On a moment band $1 \leq \beta \leq U \leq 2$, and under the frequency restriction $|t|/\sqrt{n} \leq \pi$, set

$$\delta_U := 1 - \frac{1}{U}, \quad a_U^2 := \left(1 - \frac{1}{U}\right)\left(1 + \frac{5}{3U}\right).$$

Both normalized moment coefficients are nondecreasing in U on this band, so Propositions 4.5, 4.4 and Lemma 4.7 give

$$n|\operatorname{Re}(f(u) - e^{-u^2/2})| \leq \frac{b\delta_U|t|^3}{6} + \frac{vt^4}{12}, \quad n|\operatorname{Im} f(u)| \leq \frac{ba_U|t|^3}{6},$$

whence, taking the Euclidean norm,

$$E_v(t) := \sqrt{\left(\frac{b\delta_U|t|^3}{6} + \frac{vt^4}{12}\right)^2 + \left(\frac{ba_U|t|^3}{6}\right)^2}.$$

The benefit near $\beta = 1$ is visible: both moment-dependent coordinates vanish as $U \downarrow 1$. Either prefactor is valid on its stated domain, so the smaller may be used, and the certificates select per panel.

LEAN. `realMomentRatio`, `imaginaryMomentRatioSq`, `realMomentRatio_mono`, `imaginaryMomentRatioSq_mono` in `BerryEsseen/VectorRemainder.lean`; `vectorPrefactor` and `norm_charFun_normalized_sub_gaussian_le_vector_cap` in `BerryEsseen/VectorCompression.lean`. The packaged envelopes are `scalarErrorEnvelope`, `vectorErrorEnvelope`, `modulusEnvelope`, `errorDamping` and `envelopeSmoothingBound` in `BerryEsseen/CompressedSmoothing.lean`, whose two main theorems `normalizedSum_cdf_le_scalarSmoothing` and `normalizedSum_cdf_le_vectorSmoothing` bound the actual Kolmogorov distance by an integral expression containing no unknown law. That file also proves the integrability of each envelope against the kernel, which is a genuine hypothesis of Corollary 6.7 and not something the numerics may assume. The pointwise minimum of the two envelopes is handled in `BerryEsseen/MixedSmoothing.lean`.

7.3 Choosing the caps

It remains to produce, from the cell's parameters, admissible values of v and d .

Lemma 7.4. *Let $\ell \leq b$.*

(a) *If $\beta \geq B > 0$ then $1/\sqrt{n} \leq b/B$, hence $1/n \leq (b/B)^2$. Taking $B = 2$ treats all positive sample sizes in the region $\beta \geq 2$.*

(b) *If $n \geq 20$ and $L \leq \beta$, put*

$$\rho := \min\left\{\frac{223606798}{10^9}, \frac{b}{L}\right\}, \quad v := \min\left\{\frac{1}{20}, \left(\frac{b}{L}\right)^2\right\}, \quad d := b + \rho.$$

Then $1/\sqrt{n} \leq \rho$, $1/n \leq v$ and $(\beta + 1)/\sqrt{n} \leq d$; the first of these rests on the rational inequality $20(223606798/10^9)^2 \geq 1$. Furthermore the guard $\rho T\tau \leq \pi$ implies the frequency restriction $T\tau/\sqrt{n} \leq \pi$ required by the vector prefactor.

LEAN. `large_moment_caps`, `moment_band_caps`, `minorant_argument_cap` and `reciprocal_sqrt_le_of_sample_bound` in `BerryEsseen/ParameterCaps.lean`; their use is `normalizedError_le_of_scalar_cell` and `normalizedError_le_of_moment_cell` in `BerryEsseen/ParameterCellBounds.lean`. Note $223606798/10^9 = 111803399/(5 \cdot 10^8)$; the same rational appears in Section 10 as the endpoint of each moment band.

7.4 Finite sample ranges: a geometric average

For the small sample sizes $1 \leq n \leq 19$ a different compression is used, in which the cutoff is $T_n = c\sqrt{n}$ so that the single-summand frequency $T_n s / \sqrt{n} = cs$ is independent of n . The comparison across a range $N \leq n \leq H$ then rests on the following identity.

Lemma 7.5. *For $m, g \in \mathbb{R}$ and $k \geq 1$ put $G_k(m, g) = \sum_{j=0}^{k-1} m^j g^{k-1-j}$. Then*

$$\frac{G_k(m, g)}{k} = \int_0^1 ((1-v)g + vm)^{k-1} dv. \quad (23)$$

If moreover $0 \leq m, g \leq 1$, then for $N \leq n \leq H$,

$$G_n(m, g) \leq \frac{n}{N} G_N(m, g) \leq \frac{H}{N} G_N(m, g). \quad (24)$$

Proof. For $m \neq g$, integrate the derivative of $v \mapsto ((1-v)g + vm)^k$ and use $(m-g)G_k(m, g) = m^k - g^k$; for $m = g$ both sides of (23) equal m^{k-1} . No case is excluded, including equal or zero bases. If $0 \leq m, g \leq 1$ the integrand of (23) lies in $[0, 1]$ and decreases as k increases, which gives (24). $\square$

LEAN. `BerryEsseen/FiniteGeometricAverage.lean`.

Consequently, with $u = cs$, $g = e^{-u^2/2}$, and M, E as in Section 11.1,

$$|f(u)^n - g^n| \leq \frac{H}{N} u^3 E(u) G_N(M(u), g), \quad |f(u)|^n \leq M(u)^N, \quad (25)$$

and $g^n = e^{-T_n^2 s^2 / 2}$ is exactly the comparison function that Theorem 6.6 needs.

8 Region I: small Lyapunov fractions, analytically

This region uses no numerical optimization and no certificate. It proves a *regional* constant strictly better than the global target.

Theorem 8.1. *Let μ be admissible with $\beta = \beta(\mu)$, let $n \geq 1$ and suppose $\ell = \beta/\sqrt{n} \leq \frac{1}{20}$. Then for every $x \in \mathbb{R}$,*

$$|F_n(x) - \Phi(x)| \leq \frac{183}{400} \frac{\beta}{\sqrt{n}} \quad \left(\frac{183}{400} = 0.4575 \right).$$

Consequently $N(\mu, n, x) \leq 183/400 \leq 293/625$ on this region.

We emphasize that this is a regional statement. It is *not* a bound of 0.4575 on C_{iid} : the other three regions are what force the global constant up to 0.4688.

LEAN. `BerryEsseen.normalizedError_small_fraction` and `BerryEsseen.normalizedError_le_target_of_small_fraction` in `BerryEsseen/SmallFraction.lean`.

8.1 Parameters

Put

$$r := n^{-1/2}, \quad \ell := \beta r, \quad \nu := 1 + \beta^{-1}, \quad T := \frac{2\pi}{(\beta+1)r} = \frac{2\pi\sqrt{n}}{\beta+1}, \quad \tau := \frac{3}{20}, \quad \Gamma := \sqrt{(\beta-1)(2\beta+\frac{2}{3})}.$$

Three consequences are recorded once and used throughout.

Lemma 8.2. *If $\beta \geq 1$ and $\ell \leq \frac{1}{20}$, then $n \geq 400$,*

$$(\beta+1)rT = 2\pi, \quad \pi \leq \ell T, \quad T \geq 60.$$

Proof. From $\beta \geq 1$ and $\beta/\sqrt{n} \leq \frac{1}{20}$ we get $\sqrt{n} \geq 20\beta \geq 20$, so $n \geq 400$. The first identity is the definition of T . Next $\ell T = \beta r T = 2\pi\beta/(\beta+1) \geq \pi$ because $\beta \geq 1$. Finally $r \leq \beta r \leq \frac{1}{20}$ gives $(\beta+1)r = \beta r + r \leq \frac{1}{10}$, hence $T = 2\pi/((\beta+1)r) \geq 20\pi \geq 60$. $\square$

LEAN. `small_fraction_cutoff_lower` in `BerryEsseen/SmallFractionBudget.lean`.

8.2 The cubic error coefficient and an ellipse inequality

Lemma 8.3. For $0 \leq u \leq \pi$,

$$|f(u) - e^{-u^2/2}| \leq \frac{\Gamma u^3}{6} + \frac{u^4}{8}. \quad (26)$$

Proof. Start from (12) with $c = e^{-u^2/2}$. By Lemma 4.7 the real coordinate is at most $\frac{(\beta-1)u^3}{6} + \frac{u^4}{12}$, and by Proposition 4.4 the imaginary coordinate is at most $\frac{u^3}{6} \sqrt{(\beta-1)(\beta+\frac{5}{3})}$. The elementary inequality $\sqrt{(a+b)^2 + c^2} \leq \sqrt{a^2 + c^2} + b$ ($a, b \geq 0$) peels off the quartic term, and

$$\sqrt{\left(\frac{(\beta-1)u^3}{6}\right)^2 + \left(\frac{u^3}{6} \sqrt{(\beta-1)(\beta+\frac{5}{3})}\right)^2} = \frac{u^3}{6} \sqrt{(\beta-1)^2 + (\beta-1)(\beta+\frac{5}{3})} = \frac{\Gamma u^3}{6},$$

because $(\beta-1) + (\beta+\frac{5}{3}) = 2\beta + \frac{2}{3}$. Finally $u^4/12 \leq u^4/8$. $\square$

Lemma 8.4 (Ellipse relation). For $\beta \geq 1$,

$$2\nu^2 + 3\left(\frac{\Gamma}{\beta}\right)^2 = 8, \quad \text{hence} \quad \frac{\nu}{5} + \frac{5}{48} \frac{\Gamma}{\beta} \leq \frac{7}{16}. \quad (27)$$

Proof. Writing $\nu = (\beta+1)/\beta$ and $\Gamma^2 = (\beta-1)(2\beta+\frac{2}{3})$,

$$2\frac{(\beta+1)^2}{\beta^2} + 3\frac{(\beta-1)(2\beta+\frac{2}{3})}{\beta^2} = \frac{2\beta^2 + 4\beta + 2 + 6\beta^2 - 4\beta - 2}{\beta^2} = 8.$$

For the second assertion, weighted Cauchy–Schwarz gives

$$\left(\frac{\nu}{5} + \frac{5}{48} \frac{\Gamma}{\beta}\right)^2 \leq \left(2\nu^2 + 3\left(\frac{\Gamma}{\beta}\right)^2\right) \left(\frac{1}{2 \cdot 25} + \frac{1}{3} \left(\frac{5}{48}\right)^2\right) = 8 \left(\frac{1}{50} + \frac{25}{6912}\right) = 0.18893 \dots < \left(\frac{7}{16}\right)^2 = 0.19140625. \quad \square$$

LEAN. `smallCubicCoefficient`, `sqrt_shift_square_le`, `errorBound_le_smallCubic`, `smallCubicCoefficient_ellipse`, `small_fraction_leading_coefficient` in `BerryEsseen/SmallFractionCoefficients.lean`.

8.3 Low frequencies

For $0 \leq s \leq \tau$ put $u = Ts/\sqrt{n} = Tsr$; then $(\beta+1)u = 2\pi s \leq 3\pi/10$, which is below the splice $9/2$, so the convex minorant is on its linear branch $h(v) = \frac{1}{2} - \frac{v}{10}$.

Apply Proposition 7.3 in the following concrete form. Let $m = \exp[-u^2 h((\beta+1)u)]$; then $|f(u)| \leq m$ by Theorem 5.1, and also $e^{-u^2/2} \leq m$ because $h \leq \frac{1}{2}$. Hence by (19) the geometric sum is at most nm^{n-1} , and

$$(n-1)u^2 h((\beta+1)u) \geq \frac{2}{5} n u^2 \quad \text{i.e.} \quad (n-1)h((\beta+1)u) \geq \frac{2}{5} n,$$

because $(\beta+1)u \leq 3\pi/10 \leq \frac{19}{20}$ gives $h \geq \frac{1}{2} - \frac{19}{200} = \frac{81}{200}$, and $81(n-1) \geq 80n$ for $n \geq 81$, which holds since $n \geq 400$. Since $nu^2 = T^2 s^2$, this yields

$$|f(u)^n - e^{-T^2 s^2/2}| \leq n \left(\frac{\Gamma u^3}{6} + \frac{u^4}{8} \right) e^{-2T^2 s^2/5}. \quad (28)$$

Lemma 8.5 (Kernel bound on the low band). For $0 < s \leq \tau = \frac{3}{20}$,

$$s |K(s)| \leq \frac{10}{9\pi}.$$

Proof. The two coordinates of $sK(s)$ are $s(1-s)$ and $\frac{1}{\pi} - s(1-s)q(\pi s)$. The first is increasing on $[0, \frac{1}{2}]$, so it is at most $\tau(1-\tau) = \frac{51}{400}$; the second is nonnegative and at most $\frac{1}{\pi}$ because $q \geq 0$. Hence $s^2 |K(s)|^2 \leq (\frac{51}{400})^2 + \frac{1}{\pi^2}$, and $(\frac{51}{400})^2 \leq \frac{19}{81\pi^2}$ follows from $\pi^2 \leq 10$; adding $\frac{1}{\pi^2}$ gives $\frac{100}{81\pi^2} = (\frac{10}{9\pi})^2$. $\square$

Lemma 8.6 (Finite Gaussian moments). For $b > 0$ and $R > 0$,

$$\int_0^R e^{-bs^2} ds \leq \frac{\sqrt{\pi/b}}{2}, \quad \int_0^R s^2 e^{-bs^2} ds \leq \frac{\sqrt{\pi/b}}{4b}, \quad \int_0^R s^3 e^{-bs^2} ds \leq \frac{1}{2b^2}.$$

Proof. The first is the half-line Gaussian integral. The other two follow by integrating the derivatives of se^{-bs^2} and of $-(bs^2 + 1)e^{-bs^2}/(2b^2)$ and discarding nonnegative endpoint terms. $\square$

Proposition 8.7. *The low-frequency term satisfies*

$$L := \int_0^\tau |K(s)| |\hat{\mu}_n(Ts) - e^{-T^2 s^2/2}| ds \leq \frac{5}{48} \Gamma r + \frac{7}{50} r^2.$$

Proof. Insert (26) into (28), then pair the factor s with the kernel using Lemma 8.5. With $u = Tsr$ and $nr^2 = 1$, the cubic part of the integrand becomes $\frac{10}{9\pi} \cdot \frac{\Gamma T^3 r}{6} s^2 e^{-bs^2}$ and the quartic part $\frac{10}{9\pi} \cdot \frac{T^4 r^2}{8} s^3 e^{-bs^2}$, where $b = 2T^2/5$. Apply Lemma 8.6. Since $\sqrt{\pi/b} = \sqrt{5\pi/2}/T \leq 9\pi/(10T)$ — equivalent to $\pi \geq 250/81$ — we get

$$\int_0^\tau s^2 e^{-bs^2} ds \leq \frac{\sqrt{\pi/b}}{4b} \leq \frac{9\pi}{16T^3}, \quad \int_0^\tau s^3 e^{-bs^2} ds \leq \frac{1}{2b^2} = \frac{25}{8T^4},$$

whence the cubic contribution is at most $\frac{10}{9\pi} \cdot \frac{\Gamma T^3 r}{6} \cdot \frac{9\pi}{16T^3} = \frac{5}{48} \Gamma r$ and the quartic contribution at most $\frac{10}{9\pi} \cdot \frac{T^4 r^2}{8} \cdot \frac{25}{8T^4} = \frac{125}{288\pi} r^2 \leq \frac{7}{50} r^2$. $\square$

LEAN. `small_fraction_low_integral` in `BerryEsseen/SmallFractionLow.lean`; the three ingredients are in `BerryEsseen/SmallFractionPower.lean` (power comparison), `BerryEsseen/SmallFractionKernels.lean` (Lemma 8.5), and `BerryEsseen/SmallFractionGaussian.lean` (Lemma 8.6).

8.4 Normal correction

Proposition 8.8.

$$N := \int_0^\tau \left| K(s) - \frac{i}{\pi s} \right| e^{-T^2 s^2/2} ds \leq \frac{2\pi}{5T} + \frac{\pi^3}{40T^3}.$$

Proof. By Lemma 6.8 and (17), on $s \leq \frac{3}{20}$ we get $q(\pi s)^2 \leq \pi^2 s^2/8$, hence

$$\left| K(s) - \frac{i}{\pi s} \right| = (1-s) \sqrt{1 + q(\pi s)^2} \leq 1 + \frac{\pi^2 s^2}{16},$$

using $\sqrt{1+y} \leq 1 + y/2$. Now apply Lemma 8.6 with $b = T^2/2$, for which $\sqrt{\pi/b} = \sqrt{2\pi}/T \leq 4\pi/(5T)$ (equivalent to $\pi \geq 25/8$). The constant term contributes at most $2\pi/(5T)$ and the quadratic term at most $\frac{\pi^2}{16} \cdot \frac{4\pi}{5T} \cdot \frac{1}{2T^2} = \frac{\pi^3}{40T^3}$. $\square$

LEAN. `BerryEsseen/SmallFractionNormal.lean`.

8.5 High frequencies and the tail

Split the high band at $R = 9/(4\pi)$.

Lemma 8.9. *For $\tau \leq s \leq R$ one has $s^2 h(2\pi s) \geq \frac{9}{1000}$, and for $R \leq s \leq 1$, writing $y = 1 - s$, one has $s^2 h(2\pi s) \geq \frac{3}{10} y^2$.*

Proof sketch. Writing $J(v) = v^2 h(v)$ one has $s^2 h(2\pi s) = J(2\pi s)/(4\pi^2)$. On $[\tau, R]$ the argument $2\pi s$ ranges over $[3\pi/10, 9/2]$, where J is the cubic branch $v^2/2 - v^3/10$; this increases up to $v = 10/3$ and decreases afterwards, so its minimum on the interval is at an endpoint. Evaluating at $v = 3\pi/10$ and $v = 9/2$ gives $J \geq 0.3604\dots$, and dividing by $4\pi^2$ gives $\geq 9/1000$. On $[R, 1]$ the argument lies in $[9/2, 2\pi]$, where h is the cosine branch aG with $a > \frac{5}{6}$; using $\sin(\pi y) \geq \frac{17}{20}\pi y$ one obtains $s^2 h(2\pi s) \geq \frac{3}{10} y^2$. $\square$

Proposition 8.10. *With H_1 and H_2 the parts of the high-frequency term over $[\tau, R]$ and $[R, 1]$,*

$$H_1 \leq 100 e^{-9T^2/1000}, \quad H_2 \leq \frac{50}{27T^2},$$

and moreover, for $T \geq 60$,

$$100 e^{-9T^2/1000} \leq \frac{9}{100T^2}, \quad G := \frac{1}{\pi} \int_\tau^\infty \frac{e^{-T^2 s^2/2}}{s} ds \leq \frac{9}{100T^2}.$$

Proof. By Theorem 5.1, $|\hat{\mu}_n(Ts)| \leq \exp(-T^2 s^2 h(2\pi s))$, since $(\beta + 1)Ts/\sqrt{n} = 2\pi s$. On $[\tau, R]$ the coarse kernel bound $|K(s)| \leq (4 + \pi^2)/s \leq 100$ (valid for $s \geq \frac{3}{20}$) and Lemma 8.9 give $H_1 \leq 100e^{-9T^2/1000}$. On $[R, 1]$, Lemma 6.8 together with the same cotangent estimate gives $|K(s)| \leq \frac{10}{9}y$, so

$$H_2 \leq \frac{10}{9} \int_0^{1-R} y e^{-3T^2 y^2/10} dy \leq \frac{10}{9} \cdot \frac{10}{6T^2} = \frac{50}{27T^2}.$$

For the first remainder bound, put $x = 9T^2/1000 \geq 32$ (using $T \geq 60$). From $e^x \geq x^7/7!$ we get $xe^{-x} \leq 7!/x^6 \leq 5040/32^6 \leq 81/10^7$, hence $100e^{-x} = 100(xe^{-x})/x \leq 100 \cdot \frac{81}{10^7} \cdot \frac{1000}{9T^2} = \frac{9}{100T^2}$. For the tail, $1/s \leq s/\tau^2$ on $[\tau, \infty)$ gives $G \leq e^{-T^2\tau^2/2}/(\pi T^2\tau^2)$; here $T^2\tau^2/2 \geq 40$, and $e^{-x} \leq \frac{1}{400}$ for $x \geq 40$ (from $e^x \geq x^2/2$), so with $\pi \geq 3$, $G \leq \frac{1}{400} \cdot \frac{400}{9\pi T^2} = \frac{1}{9\pi T^2} \leq \frac{9}{100T^2}$. $\square$

LEAN. `BerryEsseen/SmallFractionDecay.lean` (Lemma 8.9), `BerryEsseen/SmallFractionHigh.lean` (H_1, H_2), `BerryEsseen/SmallFractionRemainders.lean` (the two exponential remainders).

8.6 The rational budget

Proof of Theorem 8.1. Theorem 6.6 applied to μ_n with cutoff T and split $\tau = \frac{3}{20}$ gives $|F_n(x) - \Phi(x)| \leq L + H_1 + H_2 + N + G$. Collect the first-order terms: because $(\beta + 1)rT = 2\pi$ we have $\frac{2\pi}{5T} = \frac{(\beta+1)r}{5} = \frac{\nu}{5}\ell$, so by Lemma 8.4

$$\frac{5}{48}\Gamma r + \frac{2\pi}{5T} = \left(\frac{5}{48}\frac{\Gamma}{\beta} + \frac{\nu}{5}\right)\ell \leq \frac{7}{16}\ell.$$

For the remaining terms use $r \leq \ell$, $\pi \leq \ell T$ and $\ell \leq \frac{1}{20}$:

$$\frac{7}{50}r^2 \leq \frac{7}{50}\ell^2, \quad \frac{\pi^3}{40T^3} \leq \frac{\ell^3}{40} \leq \frac{\ell^2}{100}, \quad \frac{18}{100T^2} + \frac{50}{27T^2} \leq \frac{\ell^2}{4},$$

the last because $1/T^2 \leq \ell^2/\pi^2 \leq \ell^2/9$ and $(0.18 + 50/27)/9 \leq \frac{1}{4}$. The three coefficients sum to $\frac{7}{50} + \frac{1}{100} + \frac{1}{4} = \frac{2}{5}$. Therefore

$$|F_n(x) - \Phi(x)| \leq \frac{7}{16}\ell + \frac{2}{5}\ell^2 \leq \left(\frac{7}{16} + \frac{2}{5} \cdot \frac{1}{20}\right)\ell = \left(\frac{7}{16} + \frac{1}{50}\right)\ell = \frac{183}{400}\ell.$$

Lemma 2.5 with $a = \ell$ and $C = 183/400$ finishes the proof, and $183/400 < 293/625$. $\square$

LEAN. `small_fraction_total_budget` in `BerryEsseen/SmallFractionBudget.lean` is exactly the displayed rational inequality, and `normalizedError_small_fraction` in `BerryEsseen/SmallFraction.lean` assembles the five integral bounds with it.

9 Region II: large Lyapunov fractions, by Cantelli

When ℓ is large the normalization $\sqrt{n}/\beta = 1/\ell$ is small and no Fourier calculation is needed: a universal bound of $\frac{11}{20}$ on the Kolmogorov distance between *any* centred, variance-one law and the normal law suffices.

Theorem 9.1. *Let Y have mean zero and variance one, and let $F(x) = \mathbb{P}(Y \leq x)$. Then*

$$\sup_{x \in \mathbb{R}} |F(x) - \Phi(x)| \leq \frac{11}{20}.$$

Laws with atoms are included.

Proof. *One-sided Chebyshev.* For $a > 0$, Markov's inequality applied to $(Y + 1/a)^2$ gives

$$\mathbb{P}(Y \geq a) \leq \frac{\mathbb{E}(Y + 1/a)^2}{(a + 1/a)^2} = \frac{1 + a^{-2}}{(a + a^{-1})^2} = \frac{1}{1 + a^2},$$

using $\mathbb{E}(Y + 1/a)^2 = 1 + a^{-2}$ (mean zero, variance one) and $\{Y \geq a\} \subseteq \{(Y + 1/a)^2 \geq (a + 1/a)^2\}$. Applying the same to $-Y$ bounds $\mathbb{P}(Y \leq -a)$.

Upper side. For $x > 0$ this gives $F(x) \geq 1 - \frac{1}{1+x^2} = \frac{x^2}{1+x^2}$, hence $\Phi(x) - F(x) \leq \Phi(x) - \frac{x^2}{1+x^2}$. If $x \geq 1$ the right side is at most $1 - \frac{1}{2} = \frac{1}{2}$. If $0 \leq x \leq 1$, then $\varphi \leq \frac{2}{5}$ gives $\Phi(x) \leq \frac{1}{2} + \frac{2x}{5}$ and it remains to check

$$\frac{1}{2} + \frac{2x}{5} - \frac{x^2}{1+x^2} \leq \frac{11}{20} \quad (0 \leq x \leq 1).$$

Multiplying by $20(1 + x^2) > 0$, this is equivalent to $P(x) \geq 0$ where

$$P(x) := 1 - 8x + 21x^2 - 8x^3.$$

On $[0, \frac{1}{2}]$ we have $21x^2 - 8x^3 \geq 17x^2$, so $P(x) \geq 1 - 8x + 17x^2 = 17(x - \frac{4}{17})^2 + \frac{1}{17} > 0$. On $[\frac{1}{2}, 1]$ we have $21x^2 - 8x^3 \geq 13x^2$, so $P(x) \geq 1 - 8x + 13x^2 = 13(x - \frac{1}{2})^2 + 5(x - \frac{1}{2}) + \frac{1}{4} > 0$.

Lower side and negative thresholds. $F(x) - \Phi(x) \leq 1 - \Phi(x) \leq \frac{1}{2}$ for $x \geq 0$. For $x < 0$ use the lower-tail bound together with $\Phi(-x) = 1 - \Phi(x)$. At $x = 0$ use $0 \leq F(0) \leq 1$ and $\Phi(0) = \frac{1}{2}$. $\square$

LEAN. `cantelli_upper`, `cantelli_lower`, `cdf_error_le_eleven_twentieths`, `normalizedSum_cdf_error_le_eleven_twentieths` in `BerryEsseen/Cantelli.lean`; the normal density and distribution function facts ($\varphi \leq \frac{2}{5}$, monotonicity, symmetry, and the polynomial gap lemma) are in `BerryEsseen/GaussianBounds.lean`; the moment identities for the normalized sum ($\mathbb{E}S_n = 0$, $\text{Var } S_n = 1$, $S_n \in L^2$) are in `BerryEsseen/IndependentSums.lean`.

Corollary 9.2. *If μ is admissible, $n \geq 1$ and $\ell = \beta/\sqrt{n} \geq \frac{6}{5}$, then for every x ,*

$$N(\mu, n, x) \leq \frac{5}{6} \cdot \frac{11}{20} = \frac{11}{24} = 0.4583\overline{3} < \frac{293}{625}.$$

Proof. $\sqrt{n}/\beta = 1/\ell \leq \frac{5}{6}$; multiply Theorem 9.1 by it. $\square$

LEAN. `normalizedError_large_fraction` and `normalizedError_le_target_of_large_fraction` in `BerryEsseen/LargeFraction.lean`; the same file contains the conversion `upperTarget_eq_ofReal` proving $(0.4688 : \text{ENNReal}) = \text{ENNReal.ofReal}(293/625)$, which every region uses at the end.

10 Region III: 467 certified interval cells

This region covers

$$\ell \geq \frac{1}{20} \quad \text{together with} \quad (\beta \geq 2, n \geq 1) \quad \text{or} \quad (n \geq 20, \beta \text{ arbitrary}),$$

that is, everything except small fractions and small sample sizes with small moments.

Theorem 10.1. *Let μ be admissible and $x \in \mathbb{R}$.*

(a) *If $n \geq 1$, $\beta \geq 2$ and $\ell \geq \frac{1}{20}$, then $N(\mu, n, x) \leq \frac{293}{625}$.*

(b) *If $n \geq 20$ and $\ell \geq \frac{1}{20}$, then $N(\mu, n, x) \leq \frac{293}{625}$.*

LEAN. `ScalarFullHAll.large_moment_normalizedError` in `.../ScalarFullH/LargeMomentCover.lean` and `ScalarFullHAll.large_sample_normalizedError` in `.../ScalarFullH/MomentBands.lean`.

10.1 What a cell is

Fix a closed Lyapunov-fraction interval $[a, b] \subseteq [\frac{1}{20}, \frac{6}{5}]$, a third-moment floor $L \geq 1$, a sample-size floor $N_{\min} \geq 1$, a cutoff $T > 0$, a split $0 < \tau < \frac{1}{2}$ and a panel count $P \geq 1$. From these the caps v and d of Lemma 7.4 are computed, together with a moment ceiling U when the cell enables the anchored (“vector”) error prefactor. Corollary 6.7 and Proposition 7.3 then bound the Kolmogorov distance of every admissible law in the cell by

$$\mathcal{B} := \int_0^\tau |K(s)| \mathcal{E}(Ts) ds + \int_\tau^1 |K(s)| e^{-\Psi(Ts)} ds + \int_0^\tau \left| K(s) - \frac{i}{\pi s} \right| e^{-T^2 s^2/2} ds + \frac{1}{\pi} \int_\tau^\infty \frac{e^{-T^2 s^2/2}}{s} ds, \quad (29)$$

where $\Psi(t) = t^2 h(d|t)$ and the error envelope is

$$\mathcal{E}(t) := \mathcal{P}(t) e^{-(1-v)\Psi(t)} G_v(t^2/2 - \Psi(t)),$$

$\mathcal{P}$ being the scalar prefactor $\frac{b|t|^3}{6} + \frac{vt^4}{8}$, the vector prefactor E_v of Section 7, or their pointwise minimum. Note that (29) contains no unknown probability law: only the rational cell parameters.

LEAN. `ScalarFullH.GeneralCellData` and its Boolean validator `generalCellData` in `BerryEsseen/Numerics/GeneralCellCore.lean`, built on `FixedCellData/fixedCellData` in `BerryEsseen/Numerics/FixedCellCore.lean`. A `GeneralCell` is a pair of a data record and a proof that its validator returns `true`. The executable arithmetic of these validators was copied unchanged from the saved certificate core; only module and import wrappers were changed.

The validator checks, among other things,

$$\frac{1}{20} \leq a < b \leq \frac{6}{5}, \quad 1 \leq L, \quad 0 < N_{\min}, \quad 1 \leq N_{\min} \rho^2, \quad 0 < \tau < \frac{1}{2}, \quad dT\tau \leq \frac{9}{2}, \quad 0 < v < 1,$$

that the integer grid step and denominator represent exactly τ/P , and that each stored fixed-point integer really dominates (or is dominated by) the corresponding real quantity: e.g. $bT^3/6 \leq \text{cubicError}/2^{64}$, $vT^4/8 \leq \text{quarticError}/2^{64}$, $\text{cubicDecayLo}/2^{64} \leq dT^3/10 \leq \text{cubicDecayHi}/2^{64}$, $\text{quadraticDecayLo}/2^{64} \leq T^2/2$, $v \leq \text{varianceHi}/2^{64}$. When the anchor is enabled it additionally checks $L \leq U \leq 2$, $\rho T\tau \leq \pi_{\text{low}}$, and three fixed-point bounds for $b\delta_U T^3/6$, $vT^4/12$ and $b^2 a_U^2 T^6/36$. Here $\pi_{\text{low}} = 3141592653589793238/10^{18} \leq \pi$.

LEAN. The real consequences of these Boolean checks are proved in `BerryEsseen/Numerics/CellBasicFacts.lean` (cut-off and split inequalities, grid positivity, the exact relation between the rational split and the integer grid, the Gaussian quadratic coefficient), `.../CellDecayFacts.lean` (positivity of d , its relation to the stored argument scale, cubic and cosine coefficient bounds), `.../CellErrorFacts.lean` (variance cap, cubic and quartic scalar error coefficients) and `.../CellAnchorFacts.lean` (moment range, frequency restriction, separate real and imaginary coefficients). These are derived from the record checks, not supplied as extra assumptions by the generator.

10.2 Panel bounds: how endpoint data control whole integrals

Each of the three finite integrals in (29) is split into P panels $[l, r]$ with $r - l = \tau/P$ (low and normal) or into a grid on $[\tau, 1]$ (high), and each panel is bounded using only data at its endpoints. The mechanism is the following monotonicity structure.

Lemma 10.2 (The decay profile has one maximum). *Let $J(x) = x^2 h(x)$, i.e.*

$$J(x) = \begin{cases} \frac{x^2}{2} - \frac{x^3}{10}, & 0 \leq x \leq \frac{9}{2}, \\ a(1 - \cos x), & \frac{9}{2} \leq x \leq 2\pi, \\ 0, & x \geq 2\pi. \end{cases}$$

Then J increases on $[0, \frac{10}{3}]$ and decreases on $[\frac{10}{3}, \infty)$, and the branches agree at both joins. Consequently, since $\Psi(t) = J(dt)/d^2$, for $0 \leq l \leq s \leq r$,

$$\Psi(Ts) \geq \min\{\Psi(Tl), \Psi(Tr)\},$$

including for panels crossing either splice or extending into the zero branch.

LEAN. `BerryEsseen/DecayPanels.lean`; the three admissible endpoint witness modes — the value 0, the cubic formula, and a cosine-table entry — are justified in `BerryEsseen/DecayWitnesses.lean`. In the cosine mode the table argument is rounded *upwards* inside $[\frac{9}{2}, 2\pi]$, which increases the cosine and therefore decreases the certified decay, i.e. errs safely.

Proposition 10.3 (High-frequency and normal-correction panels). *Let $0 < l \leq r < 1$.*

(a) *If $\Lambda \leq \Psi(Tl)$ and $R \leq \Psi(Tr)$ then*

$$\int_l^r |K(s)| e^{-\Psi(Ts)} ds \leq (r - l) |K(l)| e^{-\min(\Lambda, R)},$$

using that $|K|$ is decreasing (Lemma 6.8).

(b) *If $q(\pi r) \leq Q$ then*

$$\int_l^r \left| K(s) - \frac{i}{\pi s} \right| e^{-T^2 s^2 / 2} ds \leq (r - l)(1 - l) \sqrt{1 + Q^2} e^{-T^2 l^2 / 2},$$

using that q is nonnegative and increasing. The case $l = 0$ is included; no value at the singular endpoint is used.

LEAN. `BerryEsseen/SmoothingPanels.lean`.

Proposition 10.4 (Low-frequency panels). *Let $0 \leq l \leq s \leq r \leq \frac{1}{2}$, let $0 < p \leq \pi$ and $0 \leq Q \leq q(\pi l)$. Then*

$$s|K(s)| \leq K_* := \sqrt{[r(1 - r)]^2 + [\frac{1}{p} - l(1 - l)Q]^2}. \quad (30)$$

If in addition $dTr \leq \frac{9}{2}$ — so that the panel stays in the cubic branch, where $(Ts)^2/2 - \Psi(Ts) = dT^3 s^3/10$ exactly — then for any $0 \leq \eta \leq dT^3 l^3/10$, any $0 < v \leq 1$, and Λ, R as above,

$$\int_l^r |K(s)| \mathcal{E}(Ts) ds \leq (r - l) K_* \mathcal{P}^\sharp(r) e^{-(1-v)\min(\Lambda, R)} G_v(\eta),$$

where $\mathcal{P}^\sharp$ is the increasing majorant of $s \mapsto s\mathcal{P}(Ts)$ on the panel:

$$\mathcal{P}^\sharp_{\text{scalar}}(s) = \frac{bT^3s^2}{6} + \frac{vT^4s^3}{8}, \quad \mathcal{P}^\sharp_{\text{vector}}(s) = s^2 \sqrt{\left(\frac{b\delta_U T^3}{6} + \frac{vT^4s}{12}\right)^2 + \left(\frac{ba_U T^3}{6}\right)^2}.$$

Proof sketch of (30). The two coordinates of $sK(s)$ are $s(1-s)$, increasing on $[0, \frac{1}{2}]$, and $\frac{1}{\pi} - s(1-s)q(\pi s)$, which is nonnegative and, being decreasing in s , is at most its value at l with the certified lower bounds $p \leq \pi$ and $Q \leq q(\pi l)$ substituted. $\square$

LEAN. `BerryEsseen/LowSmoothingPanels.lean` (scalar) and `BerryEsseen/VectorSmoothingPanels.lean` (vector); `PanelSums.lean` proves that adjacent panel integrals telescope to the complete integral, with the first and last endpoints identified explicitly.

Proposition 10.5 (Infinite normal tail). *For $\tau > 0$ and $P = T^2\tau^2/2$,*

$$\frac{1}{\pi} \int_{\tau}^{\infty} \frac{e^{-T^2s^2/2}}{s} ds \leq \frac{e^{-P}}{2\pi P},$$

using $1/s \leq s/\tau^2$ on $[\tau, \infty)$ and the elementary antiderivative of se^{-bs^2} .

LEAN. `BerryEsseen/GaussianTailBound.lean`, which also proves integrability and the form in which certified lower bounds on P and π and an upper bound on the exponential are substituted.

10.3 From a passing record to a real integral

The saved certificate stores, for each panel, a tuple of natural numbers and table references, and a Boolean check. For each of the five record kinds the development proves a theorem of the form “if the check returns `true`, then the corresponding real integral over the corresponding exact panel is at most `termUpper`/ 2^{64} ”. Table 3 lists them.

Saved check	Real conclusion	Lean theorem
<code>generalTail</code>	the infinite normal tail	<code>generalTail_sound</code>
<code>generalNormalPanel</code>	the normal-correction integral on that panel	<code>generalNormalPanel_sound</code>
<code>generalHighPanel</code>	the high-frequency modulus integral there	<code>generalHighPanel_sound</code>
<code>generalLowPanel</code> (scalar)	the scalar-error integral on that panel	<code>generalLowPanel_scalar_sound</code>
<code>generalLowPanel</code> (vector)	the vector-error integral on that panel	<code>generalLowPanel_vector_sound</code>
<code>generalLowPanel</code> (either)	the cell’s error-envelope integral	<code>generalLowPanel_sound</code>

Table 3: Record-to-integral soundness theorems; each applies to *every* passing record, and every table-entry proof and integer-to-real conversion is in its dependency closure.

Several details are worth recording, because they are exactly the places where a naive “the numbers check out” argument would be insufficient.

- The normal-panel proof uses the *upper* cotangent table entry at the *right* endpoint and the Gaussian exponential at the *left* endpoint; the high-panel proof uses both stored kernel formulas, both endpoint decay witnesses and a rounded exponential bound.
- The low-panel proof checks the lower cotangent bound, the scaled kernel norm, both decay endpoints, a lower bound for the damping difference $dT^3l^3/10$, the exponential and geometric factors, and the final product. Natural-number subtraction is converted to real subtraction only after its nonnegativity has been proved, and rounding the variance cap *upwards* weakens both damping bounds in the safe direction.
- For anchor-enabled cells the low envelope is the pointwise *minimum* of the scalar and vector envelopes; that minimum is proved integrable against the kernel and proved to bound the actual characteristic-function error, so the panel records may select either branch independently.
- `generalLowPanels_sound` requires a passing record at *every* index and proves that the grid covers exactly $[0, \tau]$; `HighGrid.lean` proves that the high grid starts at τ and ends at 1; `CellPanelSums.lean` assembles the three complete panel lists and the tail into one bound for the four-term expression (29).

LEAN. `BerryEsseen/Numerics/CellApplication.lean`, `.../CellPanelSums.lean`, `.../HighGrid.lean`, `.../IndexedPanelLists.lean`, `.../CellCertificate.lean`; `BerryEsseen/MixedSmoothing.lean` for the minimum envelope.

10.4 The budget and the covering

Definition 10.6. A *certified cell* is a `GeneralCell` together with complete lists of passing low, high and normal panel records indexed by $0, \dots, P-1$, a passing tail record, and a proof of the rational strict inequality

$$\frac{S}{2^{64}} < \frac{293}{625} \cdot a, \quad S := \sum_{\text{low}} \text{termUpper} + \sum_{\text{high}} \text{termUpper} + \sum_{\text{normal}} \text{termUpper} + \text{termUpper}_{\text{tail}}, \quad (31)$$

where a is the cell's *lower* Lyapunov-fraction endpoint.

Theorem 10.7. Let c be a certified cell with parameters $(a, b, L, N_{\min}, U)$, and let μ be admissible with $n \geq N_{\min}$, $L \leq \beta$, $a \leq \ell \leq b$, and $\beta \leq U$ if the cell enables the anchor. Then for every $x \in \mathbb{R}$,

$$N(\mu, n, x) \leq \frac{293}{625}.$$

Proof. By the panel soundness theorems and (29), $|F_n(x) - \Phi(x)| \leq \mathcal{B} \leq S/2^{64}$. By (31) this is $< \frac{293}{625}a \leq \frac{293}{625}\ell$, and Lemma 2.5 with $C = 293/625$ finishes. $\square$

LEAN. `ScalarFullH.CertifiedCell` with `CertifiedCell.normalizedError_le` in `BerryEsseen/Numerics/CertifiedCell.lean`, on top of `generalCell_certificate_normalizedError`. The hypotheses on the law are exactly its sample-size lower bound, moment range and fraction range; cutoff validity, the sample-size caps and the anchor's frequency restriction all follow from the checked cell data rather than being assumed.

Finitely many cells cover a continuum because their endpoints chain.

Lemma 10.8 (Coverage). Let $c_0, \dots, c_m$ be cells whose stored adjacency check `adjacentCells` returns *true*, i.e. $b_j = a_{j+1}$ for all j . Then every real ℓ with $a_0 \leq \ell \leq b_m$ lies in some $[a_j, b_j]$.

Proof. Induct on the list: if $\ell \leq b_0$ take $j = 0$; otherwise $\ell > b_0 = a_1$ and apply the inductive hypothesis to the tail. $\square$

LEAN. `ScalarFullH.all.adjacentCells_cover` and `certified_cover_normalizedError` in `.../ScalarFullH/Coverage.lean`. This proves coverage of the continuous parameter interval, not merely of the stored endpoints.

10.5 The five coverings

The saved data consist of 467 cells, `cell10` through `cell1466`, grouped into five chains. Their parameters are summarized in Table 4.

Third-moment range	Sample sizes	Cells	Lean indices
$\beta \geq 2$	$n \geq 1$	137	<code>cell10–cell1136</code>
$1 \leq \beta \leq \frac{11}{10}$	$n \geq 20$	76	<code>cell1137–cell1212</code>
$\frac{11}{10} \leq \beta \leq \frac{9}{4}$	$n \geq 20$	83	<code>cell1213–cell1295</code>
$\frac{9}{4} \leq \beta \leq \frac{3}{2}$	$n \geq 20$	82	<code>cell1296–cell1377</code>
$\frac{3}{2} \leq \beta \leq 2$	$n \geq 20$	89	<code>cell1378–cell1466</code>
<i>total</i>		467	

Table 4: The five chains of certified interval cells. The counts were verified by listing `BerryEsseen/Certificates/ScalarFullH/`, which contains 467 files `Cellk.lean` together with `Data.lean`, `Complete.lean`, `Coverage.lean`, `LargeMomentCover.lean`, `MomentBands.lean` and `import_manifest.json`.

The large-moment chain. The first chain has $a_0 = \frac{1}{20}$, $b_{136} = \frac{6}{5}$, moment floor $L = 2$, $N_{\min} = 1$ and the anchor disabled. By Lemma 10.8 it covers every $\ell \in [\frac{1}{20}, \frac{6}{5}]$ for laws with $\beta \geq 2$; and $\ell > \frac{6}{5}$ is Corollary 9.2. This proves Theorem 10.1(a).

The four moment bands. Each band has $N_{\min} = 20$, a common moment floor L and ceiling U taken from Table 4, the anchor enabled, and first endpoint $a \leq \frac{1}{20}$. The crucial point is the *last* endpoint: because $n \geq 20$ implies

$$\frac{1}{\sqrt{n}} \leq \frac{111803399}{500000000} = 0.223606798, \quad (32)$$

we have $\ell = \beta/\sqrt{n} \leq U \cdot \frac{111803399}{500000000}$, and the Lean band lemma requires exactly the hypothesis $U \cdot \frac{111803399}{500000000} \leq b_{\text{last}}$, i.e. that the chain's last endpoint reaches that rational bound. (The project's notes record that each band's final endpoint *equals* it.) So no large-sample tail is omitted: for $n \geq 20$ and $L \leq \beta \leq U$ the actual fraction always lies inside the chain. Splitting $1 \leq \beta \leq 2$ at $\frac{11}{10}$, $\frac{5}{4}$, $\frac{3}{2}$ and appending the large-moment chain for $\beta > 2$ proves Theorem 10.1(b).

LEAN. `moment_cover_normalizedError` in `.../ScalarFullH/Coverage.lean` (the band argument, including (32) through `reciprocal_sqrt_le_of_sample_bound`); `moment_band1_normalizedError` through `moment_band4_normalizedError`, `bounded_moment_large_sample_normalizedError` and `large_sample_normalizedError` in `.../MomentBands.lean`; `large_moment_middle_normalizedError` and `large_moment_normalizedError` in `.../LargeMomentCover.lean`.

Remark 10.9 (Scale and size). The scalar cells use the fixed-point scale 2^{64} (`s64` = 18446744073709551616). A single cell file is a few hundred lines of integer data with one `decide +kernel` per record; for example `Cell10.lean` has 795 lines, 128 low panels, 128 high panels, 128 normal panels, one tail, and the final budget line

```
theorem cell10Bound :
  (sum0 : Rat)/(s64 : Rat) < (293/625 : Rat) * cell10.data.ellLower :=
  by decide +kernel
```

Its recorded parameters are $a = \frac{1}{20}$, $b = \frac{41}{800} = 0.05125$, $L = 2$, $T = \frac{326460989}{4000000} = 81.615\dots$, $\tau = \frac{1635671}{20000000} = 0.0817835\dots$, $P = 128$, $N_{\min} = 1$, anchor disabled.

11 Region IV: 174 certified finite-sample cells

The remaining parameters are $1 \leq n \leq 19$ with $1 \leq \beta < 2$ and $\ell < \frac{6}{5}$. Here a different reduction is used, because for such small n the compression of Section 7 is wasteful: instead the cutoff is chosen proportional to $\sqrt{n}$, which freezes the single-summand frequency.

Theorem 11.1. *Let μ be admissible with $\beta < 2$ and let $1 \leq n < 20$. Then $N(\mu, n, x) \leq \frac{293}{625}$ for every $x \in \mathbb{R}$.*

LEAN. `FiniteParameterBlocks.finite_normalizedError` in `.../FiniteParameterBlocks/Coverage.lean`.

11.1 The cell reduction

A finite cell specifies integers $1 \leq N \leq n \leq H$, a moment interval $1 \leq L \leq \beta \leq B$, a cutoff constant $c > 0$ and a split $0 < \tau \leq \frac{1}{2}$, and sets

$$T_n := c\sqrt{n}, \quad \text{so that} \quad \frac{T_n s}{\sqrt{n}} = cs \quad \text{is independent of } n.$$

Choose functions M and E with

$$|f(u)| \leq M(u) \leq 1 \quad (0 < u < c), \quad |f(u) - e^{-u^2/2}| \leq u^3 E(u), \quad E \geq 0 \quad (0 < u < c\tau).$$

All moment-dependent bounds are evaluated at the *upper* endpoint B ; their monotonicity in β justifies the substitution.

The two error coefficients used by the records are the classical

$$E_{\text{classical}}(u) = \frac{B}{6} + \frac{u}{8}$$

and, for $0 \leq u \leq \pi$, the anchored

$$E_{\text{anchor}}(u) = \sqrt{\left(\frac{B-1}{6} + u Q(u^2)\right)^2 + \frac{(B-1)(B+\frac{5}{3})}{36}},$$

where Q is the explicit rational polynomial with $0 \leq e^{-u^2/2} - \cos u \leq u^4 Q(u^2)$ of Lemma 4.7. The anchored form is exactly Propositions 4.5, 4.4 combined in the Euclidean norm, with the nonnegative factor u^3 taken outside the square root; the case $u = 0$ is included.

The available smooth modulus branches are

$$M_{\text{line}}(u) = \sqrt{1 - u^2 + \frac{B+1}{5}u^3} \quad ((B+1)u \leq \frac{9}{2}),$$

$$M_{\text{cos}}(u) = \sqrt{1 - \frac{2a(1 - \cos((B+1)u))}{(B+1)^2}} \quad (\frac{9}{2} \leq (B+1)u \leq 2\pi), \quad a = \frac{81}{80(1 - \cos \frac{9}{2})},$$

the constant bound $M \equiv 1$, and an anchored branch

$$M_{\text{anchor}}(u) = \sqrt{(|\cos u| + \frac{(B-1)u^3}{6})^2 + I(u)^2},$$

with I any of the three proved imaginary-part bounds of Proposition 4.4. To differentiate the anchored branch on a panel, the sign of $\cos u$ is fixed and certified on the whole panel, the expression under the square root is checked to be at most one, and strict positivity of the radicand is checked (square-root derivatives require it).

LEAN. `BerryEsseen/FiniteSampleBounds.lean` (the two error coefficients and their application to the actual i.i.d. law), `BerryEsseen/FiniteModulus.lean` (the branch inequalities with all domain conditions explicit), `BerryEsseen/GaussianPolynomial.lean` (Q).

Combining (25) with the rational kernel envelopes

$$K_0(s) = \sqrt{[s(1-s)]^2 + [\frac{1}{\pi} - s(1-s)q_-(\pi s)]^2},$$

$$K_R(s) = (1-s)\sqrt{1 + q_+(\pi(1-s))^2}, \quad K_C(s) = (1-s)\sqrt{1 + q_+(\pi s)^2},$$

for which $s|K(s)| \leq K_0(s)$ on $(0, \frac{1}{2}]$, $|K(s)| \leq K_R(s)$ on $[\frac{1}{2}, 1)$ and $|K(s) - \frac{i}{\pi s}| \leq K_C(s)$ on $(0, \frac{1}{2}]$ (Lemmas 6.8 and 6.9), gives the following theorem. Multiplication by s in K_0 is what removes the low-frequency singularity.

Theorem 11.2 (Finite-sample smoothing bound). *With the above data define*

$$\mathcal{L}(s) = \frac{H}{N} c^3 s^2 K_0(s) E(cs) G_N(M(cs), e^{-c^2 s^2/2}), \quad \mathcal{V}(s) = K_C(s) e^{-c^2 N s^2/2},$$

$$\mathcal{U}(s) = \begin{cases} K_0(s) M(cs)^N / s, & s \leq \frac{1}{2}, \\ K_R(s) M(cs)^N, & s > \frac{1}{2}, \end{cases} \quad \mathcal{J} = \frac{e^{-c^2 N \tau^2/2}}{\pi c^2 N \tau^2}.$$

Then for every admissible law with $\beta \leq B$, every $N \leq n \leq H$ and every $x \in \mathbb{R}$,

$$|F_n(x) - \Phi(x)| \leq \mathcal{B} := \int_0^\tau \mathcal{L} + \int_\tau^1 \mathcal{U} + \int_0^\tau \mathcal{V} + \mathcal{J}.$$

Consequently, if $a \geq 0$ satisfies $H \leq (aL)^2$ and $a\mathcal{B} \leq \frac{293}{625}$, then $N(\mu, n, x) \leq \frac{293}{625}$ throughout the cell, because $\sqrt{n}/\beta \leq \sqrt{H}/L \leq a$.

LEAN. `BerryEsseen.cdf_le_finiteSmoothingBound` and `normalizedError_le_finite_budget` in `BerryEsseen/FiniteSmoothing.lean`. The three integrability hypotheses are explicit premises of the theorem: numerical data must discharge them, not merely approximate the integrals. The Gaussian factor decreases in n , so evaluating the normal correction and the tail at N bounds them for all $n \geq N$; the tail estimate itself is

$$\int_\tau^\infty \frac{e^{-c^2 n s^2/2}}{\pi s} ds \leq \frac{e^{-c^2 N \tau^2/2}}{\pi c^2 N \tau^2}$$

from `BerryEsseen/FiniteTails` (see Section 11.6).

11.2 What a normalized Taylor panel proves

The three integrals of Theorem 11.2 are bounded panel by panel using a signed fourth-order Taylor estimate. This is the single most-used real-analysis lemma of the numerical part.

Theorem 11.3 (Taylor panel bound). *Let F be four times continuously differentiable on $[m-h, m+h]$, $h \geq 0$, and suppose*

$$F(m) \leq U_0, \quad F''(m) \leq U_2, \quad F''''(s) \leq U_4 \quad \text{for all } s \in [m-h, m+h].$$

Then

$$\int_{m-h}^{m+h} F(s) ds \leq 2hU_0 + \frac{h^3}{3}U_2 + \frac{h^5}{60}U_4. \quad (33)$$

The bounds U_0, U_2, U_4 may be negative.

Proof sketch. Taylor's theorem with the integral (or Lagrange) remainder at the midpoint gives $F(s) \leq \sum_{k \leq 3} \frac{F^{(k)}(m)}{k!}(s-m)^k + \frac{U_4}{24}(s-m)^4$ for s in the panel. The odd powers integrate to zero over the symmetric interval, $\int (s-m)^2 = \frac{2}{3}h^3$ and $\int (s-m)^4 = \frac{2}{5}h^5$, and the fourth-order remainder has a nonnegative weight on both sides of m . $\square$

In the certificates the rescaled function $G(t) = F(m+ht)$ is used, and the stored quantities are enclosures of the normalized coefficients $G^{(k)}(t)/k!$, which already include the factor h^k . In those terms (33) reads

$$\int_{m-h}^{m+h} F(s) ds \leq 2h \left(u_0 + \frac{u_2}{3} + \frac{u_4}{5} \right), \quad (34)$$

where $u_0 \geq G(0)$, $u_2 \geq G''(0)/2!$, and — crucially — $u_4 \geq G^{(4)}(t)/4!$ for every $t \in [-1, 1]$, an enclosure on the whole panel and not a sampling of the derivative.

LEAN. `taylor_remainder_on_segment`, `cubic_upper_of_fourth_deriv_le`, `integral_centered_pow`, `midpoint_integral_fourth_bound` in `BerryEsseen/TaylorPanel.lean`; `normalized_midpoint_integral_bound` and `integral_le_normalized_jet_bound` in `BerryEsseen/Numerics/TaylorPanelSoundness.lean`, which also connects the scale-2³² integer interval endpoints to the real coefficient bounds.

11.3 Endpoint (Darboux) panels for the high band

On part of the high band a cruder but cheaper estimate is used, in which only the two endpoints of the panel matter. This works because the line-branch radicand is a cubic with a single interior minimum.

Lemma 11.4. *For $B \geq 1$ let $P_B(u) = 1 - u^2 + \frac{B+1}{5}u^3$ and $m_B(u) = \sqrt{P_B(u)}$. Then*

$$P'_B(u) = \frac{u}{5}(3(B+1)u - 10),$$

so the unique interior critical point $u_* = \frac{10}{3(B+1)}$ is a minimum, and

$$P_B(u) - \left(1 - \frac{100}{27(B+1)^2}\right) = \frac{(3(B+1)u - 10)^2(3(B+1)u + 5)}{135(B+1)^2} \geq 0 \quad (u \geq 0).$$

Consequently, on any panel $[l, r]$ with $(B+1)cr \leq \frac{9}{2}$,

$$m_B(cs) \leq v := \max\{m_B(cl), m_B(cr)\} \quad (l \leq s \leq r).$$

LEAN. `BerryEsseen/FiniteRadicandRange.lean`.

Proposition 11.5 (Endpoint panel bound). *Let $0 < l \leq s \leq r \leq 1$, $(B+1)cr \leq \frac{9}{2}$, $1 \leq N \leq n$, $T = c\sqrt{n}$. Then*

$$\int_l^r |K(s)| |\hat{\mu}_n(Ts)| ds \leq (r-l) \left(1 - \frac{l+r}{2}\right) \sqrt{1 + \left(\frac{1+l}{\pi l}\right)^2} v^N. \quad (35)$$

Proof. By (25) and Lemma 11.4, $|\hat{\mu}_n(Ts)| \leq m_B(cs)^N \leq v^N$. By Lemma 6.8 and Lemma 6.10,

$$|K(s)| = (1-s) \sqrt{1 + q(\pi(1-s))^2} \leq (1-s) \sqrt{1 + \left(\frac{1+s}{\pi s}\right)^2} \leq (1-s) \sqrt{1 + \left(\frac{1+l}{\pi l}\right)^2},$$

because $(1+s)/(\pi s)$ decreases. Integrating the remaining affine factor exactly, $\int_l^r (1-s) ds = (r-l)(1 - \frac{l+r}{2})$, gives (35). $\square$

The certificate for such a panel is entirely rational. With $p = 13493037704/2^{32} < \pi$ it proposes nonnegative rationals

$$M \geq \max\{\sqrt{P_B(cl)}, \sqrt{P_B(cr)}\}, \quad \Lambda \geq \sqrt{1 + ((1+l)/(pl))^2},$$

which Lean verifies by squaring, checks the parameter and branch conditions, and checks the rational inequality $(r-l)(1 - \frac{l+r}{2})\Lambda M^N \leq U$. The conclusion is that the actual integral is at most U , uniformly over every admissible law with $\beta \leq B$ and every $n \geq N$.

LEAN. `BerryEsseen/FiniteDarboux.lean` (the analytic statements) and `BerryEsseen/Numerics/DarbouxCertificate.lean` (the checker and its soundness theorems `Panel.sound`, `CertifiedPanel.sound`). The 458 instances are collected in `BerryEsseen/Certificates/FiniteDarbouxPanels/Complete.lean`; their manifest records 174 rows, 4034 source panels and 458 generated entries.

11.4 Normal-correction panels

For a cell with $N \leq n \leq H$, $T = c\sqrt{n}$ and $0 < \tau \leq \frac{1}{2}$, the normal-correction term is bounded by

$$\int_0^\tau (1-s)\sqrt{1+q_+(\pi s)^2} e^{-c^2 N s^2/2} ds,$$

with q_+ the rational upper bound of Lemma 6.9, whose denominator is positive on $[0, \frac{\pi}{2}]$. Increasing n decreases the Gaussian factor, so a bound at N applies for all $n \geq N$.

Each panel proof supplies the hypotheses of (34) as follows.

1. Every arithmetic node carries an exact function of t , a proof that it is four times continuously differentiable at each point of its domain, and five interval enclosures of its normalized derivatives.
2. Constants use the 693 exact scalar enclosures of Section 12.3; arithmetic uses the proved Taylor recurrences; inverses and square roots carry checked domain guards; exponentials use certified seeds.
3. One evaluation encloses the derivatives at $t = 0$ and a second encloses them throughout $[-1, 1]$, and Lean checks that the two evaluations represent the *same* function.
4. Lean proves an explicit identity between that function and the displayed normal envelope after the affine change of variable, unfolding the rational polynomials and simplifying the exact arithmetic.
5. (34) yields a rational panel bound, and the analytic envelope inequality transfers it to the actual Prawitz correction integral.

There are 669 normal panels across the 174 cells. A cell's list is checked to begin at 0, to have adjacent endpoints, to end exactly at its split, and to use the same cutoff and sample-size floor throughout; the endpoint check also verifies that all intervals lie in the required half-period.

LEAN. `BerryEsseen/FiniteNormalPanels.lean`, `BerryEsseen/FiniteNormalCover.lean`, `BerryEsseen/Certificates/FiniteNormalPanels/` (669 files `Paneli_j.lean` plus `Complete.lean` and `manifest.json`).

11.5 High- and low-frequency Taylor panels

The high-frequency Taylor panels bound

$$\int_l^r K_R(s) M_B(cs)^N ds \quad \text{on } [l, r] \subseteq [\tfrac{1}{2}, 1],$$

and on a left-half panel the kernel factor is instead $K_0(s)/s$ with $0 < l \leq r \leq \frac{1}{2}$, whose denominator and sine-polynomial domain are checked throughout the panel. The branch conditions are checked as inequalities valid on the whole panel: the line branch checks $(B+1)cr \leq \frac{9}{2}$, the cosine branch checks both $(B+1)cl \geq \frac{9}{2}$ and $(B+1)cr \leq 2\pi$. Continuity of the kernel envelope and of the modulus gives integrability of the target, including at the endpoint $s = 1$. Anchored panels additionally use whole-panel cosine-sign and squared-modulus bounds to prove the sign and the bound $M \leq 1$ at every real point of the panel; no sign or domain condition is inferred from a sampled function value.

The low-frequency Taylor panels bound $\int \mathcal{L}$ using the previously proved sample-interval comparison (24) and either the anchored or the classical error coefficient. The special case $N = 1$ has geometric factor one and therefore does not require an unused modulus branch.

Counts: 1806 high-frequency Taylor panels and 1101 low-frequency Taylor panels, verified against the two manifests (`generated_entries=expected_entries` with `complete_cover=true` in both).

LEAN. `BerryEsseen/FiniteHighPanels.lean`, `BerryEsseen/FiniteLowPanels.lean`, `BerryEsseen/FiniteHighCover.lean`, `BerryEsseen/FiniteLowCover.lean`; `BerryEsseen/Certificates/FiniteHighTaylorPanels/` (91 batch files) and `.../FiniteLowTaylorPanels/` (111 batch files); `Numerics/JetPanelRange.lean` for the affine-inverse rule that turns a proved whole-panel zeroth coefficient into a pointwise bound.

11.6 Gaussian tails

For $n \geq N > 0$, $c > 0$, $\tau > 0$, put $Q = c^2 N \tau^2$, a positive rational in every saved cell, so that the exponent is the exact rational $-Q/2$. The checker verifies: the sample floor, cutoff and split are positive; the exponential seed's input interval contains $-Q/2$; and its certified output upper endpoint $\mathcal{E}$ satisfies $\mathcal{E}/(\pi_{\text{low}} Q) \leq U$ with $\pi_{\text{low}} = 13493037704/2^{32} \leq \pi$. Monotonicity of the exponential and division by positive denominators then prove the real tail bound. All 174 cells have checked tail certificates, and each applies to every sample size above the recorded floor.

LEAN. `BerryEsseen/Numerics/FiniteTailCertificate.lean` (`tailCheck`, `CertifiedTail.envelope_bound`, `CertifiedTail.sound`) and `BerryEsseen/Certificates/FiniteTails/Complete.lean`.

11.7 Cell and block assembly

Definition 11.6. A *certified finite cell* bundles a certified low cell, high cell, normal cell and tail, all with the same N , H , B , c , τ , together with rationals L (moment floor), a (normalization factor) and the constant $\frac{293}{625}$, and a Boolean check that includes

$$0 < N, \quad 0 < c, \quad 0 < \tau \leq 1, \quad 0 < L \leq B, \quad a \geq 0, \quad H \leq (aL)^2, \quad 0 \leq \mathcal{U}, \quad a\mathcal{U} \leq \frac{293}{625},$$

where $\mathcal{U}$ is the sum of the four stored rational upper bounds.

LEAN. `finiteCellCheck`, `CertifiedFiniteCell` and `CertifiedFiniteCell.sound` in `BerryEsseen/FiniteCellCertificate.lean`. The soundness proof combines Theorem 11.2 with $\sqrt{H}/L \leq a$ (obtained from $H \leq (aL)^2$ by taking square roots) and Lemma 2.5.

The 174 certified finite cells are grouped into 17 blocks by sample-size range, each block chaining moment intervals so that adjacent cells meet exactly. Table 5 lists them.

Sample sizes $[N, H]$	Moment ceiling	Cells
[1, 1]	$\frac{12001}{10000}$	2
[2, 2]	$\frac{16971}{10000}$	9
[3, 3]	2	13
[4, 4]	2	14
[5, 5]	2	14
[6, 6]	2	13
[7, 7]	2	11
[8, 8]	2	9
[9, 9]	2	9
[10, 10]	2	9
[11, 11]	2	9
[12, 12]	2	9
[13, 13]	2	9
[14, 14]	2	9
[15, 16]	2	14
[17, 18]	2	13
[19, 19]	2	8
<i>total</i>		174

Table 5: The 17 finite-sample parameter blocks. The counts were verified by reading `BerryEsseen/Certificates/FiniteParameterBlocks/Block $k$ .lean` and counting the `FiniteCompleteCells.cell $j$` references in each.

Proof of Theorem 11.1. If $\ell \geq \frac{6}{5}$ apply Corollary 9.2. Otherwise $\beta < 2$ and $\ell < \frac{6}{5}$; the elementary lemma

$$\beta < 2 \wedge \frac{\beta}{\sqrt{n}} < \frac{6}{5} \implies \beta < \begin{cases} \frac{12001}{10000}, & n = 1, \\ \frac{16971}{10000}, & n = 2, \\ 2, & n \geq 3, \end{cases}$$

(for $n = 1$ this is immediate; for $n = 2$ it uses $\sqrt{2} \leq \frac{28285}{20000}$) reduces the moment range to the block's ceiling. Since $1 \leq n \leq 19$, one of the 17 blocks applies; its chained moment intervals cover $[1, \text{ceiling})$ by the same argument as Lemma 10.8, and Definition 11.6 gives the bound $\frac{293}{625}$. $\square$

LEAN. `finiteMomentCap` and `lt_finiteMomentCap` in `BerryEsseen/FiniteMomentCap.lean`; `FiniteParameterCover.lean` for the adjacency of moment intervals; `FiniteParameterBlocks.finite_small_moment_bound` and `finite_normalizedError` in `.../FiniteParameterBlocks/Coverage.lean`. The Lean proof performs an `interval_cases n` over $1 \leq n \leq 19$, dispatching $n = 16$ to the $[15, 16]$ block and $n = 18$ to the $[17, 18]$ block.

Remark 11.7 (The tightest saved budget). The manifest of `FiniteCompleteCells` records that the largest normalized budget among the 174 cells is at row 48, namely

$$\frac{41323819126424383349609375}{88230205055486500383227904} = 0.4683636301245874\dots < \frac{293}{625} = 0.4688.$$

That cell has $N = H = 5$, $B = \frac{3^{18}}{5^{12}} = 1.58672\dots$, $c = \frac{239773}{100000}$, $\tau = \frac{36843}{100000}$, $L = \frac{3^{15}}{5^{10}} = 1.46932\dots$, $a = \frac{93787488623046875}{61628086298345472} = 1.52185\dots$, and nine low-frequency panels. The margin to the target is about 0.0004, so the budgets are tight but not marginal.

12 From integers to real integrals

The numerical part of the proof consists of exact integer and rational inequalities. Its correctness rests on a chain of soundness theorems, each of which says that a particular *passing check* implies a particular *real* statement. A list of passing integer inequalities is never treated as a proof of a probability bound on its own.

12.1 Interval arithmetic with outward rounding

Definition 12.1. Fix a scale S . An integer pair $[\alpha, \beta]$ *encloses* a real x when $\alpha \leq Sx \leq \beta$. Two scales occur:

$$S = 2^{32} = 4294967296 \quad (\text{the Taylor-jet records of the finite-sample cells}),$$

$$S = 2^{64} = 18446744073709551616 \quad (\text{the interval cells of Section 10}).$$

Every arithmetic operation rounds outwards. Addition and negation are exact on endpoints; multiplication takes the smallest and largest of the four endpoint products, rounding the former down and the latter up after division by S . Square-root witnesses are verified by squaring integers; reciprocals carry a nonzero-denominator guard.

LEAN. `BerryEsseen/Numerics/JetCore.lean` defines the interval type and the executable operations at scale 2^{32} ; `BerryEsseen/Numerics/IntervalSoundness.lean` proves that each rule preserves real enclosures, including for intervals with endpoints of different signs (`Contains.add`, `Contains.neg`, `Contains.sub`, `Contains.mul`, `Contains.scaleNat`, `Contains.divConst`, and the outward-rounding lemmas `floorScale_bound`, `ceilScale_bound`).

12.2 Taylor jets: from an expression to derivative bounds

A *jet* stores the five normalized coefficients $c_j = F^{(j)}(x)/j!$, $j = 0, \dots, 4$, as intervals. Products use ordinary convolution $(fg)_k = \sum_{j \leq k} f_j g_{k-j}$, and the other operations follow from identities that determine the next coefficient from earlier ones:

Operation	Determining identity
$y = 1/f$	$fy = 1$
$y = \sqrt{f}$	$y^2 = f$
$y = e^f$	$y' = f'y$
$(s, c) = (\sin f, \cos f)$	$s' = f'c, c' = -f's$

The Lean proofs connect these recurrences to *actual derivatives* of the functions involved, and they require a nonzero denominator or a positive square-root argument where the identity needs one; the executable guards establish these conditions. An exponential or trigonometric node additionally needs a valid interval for its *zeroth* coefficient, which the recurrence cannot supply; that premise is explicit, and is discharged by the seed certificates of Sections 12.4 and 12.5.

LEAN. `BerryEsseen/Numerics/DifferentialJet.lean`, `.../InverseJet.lean`, `.../SquareRootJet.lean`, `.../ExponentialJet.lean`, `.../TrigonometricJet.lean`, `.../JetSoundness.lean`, and the whole-domain form `EnclosesOn` in `.../EnclosesOn.lean`, whose constructors (`EnclosesOn.add`, `.mul`, `.inv`, `.sqrt`, `.exp`, `.sincos`, `.affine`, `.constant`, `.refine_radican`) carry the domain conditions as hypotheses. The affine input rule and the rule for replacing a zeroth interval by a separately proved tighter enclosure are in `.../JetInputs.lean`.

12.3 Scalar constants keep their exact real meanings

A rational interval alone does not identify which real constant it encloses: a rounded interval may enclose several different rationals. The construction therefore records the exact real value *in the type*:

```
structure ScalarEnclosure (value : Real) where
  interval : Interval
  contains : interval.Contains value
```

The parameter `value` is part of the mathematical statement, so it cannot change when an arithmetic operation or an interval widening is applied. Each constructor produces an enclosure of the corresponding exact real operation, and the theorem `encloses_constant` initializes a constant Taylor jet from such an enclosure.

The finite Taylor formulas use 69,759 constant occurrences across 3,576 panels. A complete replay reduces them to 647 distinct constant records plus 46 supporting scalar operations, i.e. 693 scalar enclosures in all, namely 506 rational enclosures, one π enclosure, one cosine, one subtraction, two reciprocals, 152 products and 30 square roots. Among the constants actually formed are

$$c\sqrt{N}, \quad \sqrt{(B-1)(B+\frac{5}{3})}, \quad \sqrt{B-1}, \quad \frac{1}{\pi}, \quad \frac{81}{80(1-\cos\frac{9}{2})},$$

together with the rational polynomial coefficients and the sample and moment endpoints. The square-root rule permits the value zero, which is necessary for $\sqrt{B-1}$ when $B=1$; trigonometric constants use the already proved endpoint and range certificates.

LEAN. `BerryEsseen/Numerics/ScalarEnclosure.lean` (definitions and soundness rules: `rational`, `pi`, `add`, `neg`, `sub`, `mul`, `divNat`, `inv`, `sqrt`, `exp`, `sin`, `cos`, `widen`) and `BerryEsseen/Certificates/FiniteScalarConstants/Complete.lean` (the 693 checked records; the directory has seven batch files and a manifest recording `generated_entries=693`). The public data file `data/finite_scalar_meanings.json` records all scalar nodes, all 69,759 occurrence mappings and the 3,576 sets of coefficient targets; it is untrusted input to certificate construction.

12.4 Exponential seeds

The finite-sample programs contain 3,514 exponential operations. For each, the starting interval for $e^{x/S}$ ($S=2^{32}$) is certified as follows. Choose $k \geq 0$ and put $r = x/(S2^k)$, requiring $|r| \leq 1$. For any $n \geq 1$, Taylor's theorem gives

$$\left| e^r - \sum_{j < n} \frac{r^j}{j!} \right| \leq R := \frac{2|r|^n}{n!}.$$

Both the polynomial P and R are exact rationals, and a proposed seed $[\alpha, \beta]$ is accepted only if $\alpha \leq S(P-R)$ and $S(P+R) \leq \beta$, which implies $\alpha \leq Se^r \leq \beta$. The original argument is recovered by squaring the interval outwards k times using the proved integer multiplication rule, which encloses $(e^r)^{2^k} = e^{x/S}$; the final proposed interval must *contain* the computed one, so harmless widening is allowed but unjustified narrowing is not. Two shortcuts have separate proofs: $x=0$ gives $e^0=1$, and $x/S \leq -32$ gives $e^{x/S} \leq e^{-32} \leq 2^{-32}$ (using $e \geq 2$), so $[0, 1]$ is valid. Finally, for an input interval $[l, u]$ one certifies both endpoints and uses monotonicity of $\exp$ to cover every real y with $l \leq Sy \leq u$; the lower bound may also be raised to 0 since $\exp > 0$. No floating-point exponential occurs anywhere in the check.

LEAN. `BerryEsseen/Numerics/ExpSeedCore.lean` (executable `check`), `.../ExpSeedSoundness.lean` (`expIntervalCheck_sound`), `.../CertifiedExponentialJet.lean` (`Encloses.exp_certified`, which requires the checked input interval to *equal* the actual zeroth jet coefficient), and `BerryEsseen/Certificates/FiniteExpSeeds/` (36 batch files plus `Complete.lean`; manifest: 36 sources, 3,576 panels, 3,514 unique entries).

12.5 Trigonometric seeds

The frozen programs contain 4,248 paired sine/cosine calls with input intervals inside $[0, 2\pi]$. Each certificate proves bounds for every real input in its interval, including intervals crossing a critical point. The proof has two parts.

Endpoint values. The fixed interval $[13493037704, 13493037705]$ encloses $S\pi$ at $S = 2^{32}$, which follows from mathlib’s proved decimal bounds on π . Choosing a quadrant witness $k \in \{0, 1, 2, 3, 4\}$, outward integer arithmetic encloses $z = x/S - k\pi/2$, and the checker verifies that this interval lies in $[-1, 1]$; it need not prove that k is the nearest quadrant, since only the inclusion and the trigonometric identities are used. For the reduced argument,

$$C(z) = 1 - \frac{z^2}{2!} + \cdots + \frac{z^{12}}{12!}, \quad S(z) = z - \frac{z^3}{3!} + \cdots + \frac{z^{13}}{13!},$$

and since the next Taylor coefficient vanishes in each case,

$$|\cos z - C(z)| \leq \frac{|z|^{14}}{14!}, \quad |\sin z - S(z)| \leq \frac{|z|^{15}}{15!},$$

both smaller than 2^{-32} for $|z| \leq 1$. Each polynomial is evaluated by outward Horner arithmetic and each endpoint widened by one integer unit. The original values are reconstructed by the quadrant identities

$k \bmod 4$	$\sin(x/S)$	$\cos(x/S)$
0	$\sin z$	$\cos z$
1	$\cos z$	$-\sin z$
2	$-\sin z$	$-\cos z$
3	$-\cos z$	$\sin z$

with a separate shortcut at $x = 0$.

Values between the endpoints. A continuous differentiable function on a closed interval attains its extrema at an endpoint or at an interior critical point; the Lean proof uses the extreme-value theorem and Fermat’s theorem, and does *not* infer a range from sampled values. Inside $(0, 2\pi)$ the relevant critical data are

$$\sin : \pi/2 \mapsto 1, \quad 3\pi/2 \mapsto -1; \quad \cos : \pi \mapsto -1.$$

The checker uses outward intervals for these locations: if a critical-location interval overlaps the input interval, the output must include the corresponding value ± 1 . An overlap may be conservative but cannot cause a critical point to be missed. Endpoint enclosures may also be intersected with $[-1, 1]$.

LEAN. BerryEsseen/Numerics/TrigPolynomialSoundness.lean, .../TrigEndpointSoundness.lean, BerryEsseen/TrigonometricRange.lean (critical-point classification and range theorem), .../TrigSeedCore.lean, .../TrigSeedSoundness.lean (trigIntervalCheck_sound), .../CertifiedTrigonometricJet.lean (Encloses.sincos_certified), and BerryEsseen/Certificates/FiniteTrigSeeds/ (43 batch files; manifest: 4,248 unique entries).

12.6 Radicand refinements

The finite Taylor programs contain 2,884 operations that replace the interval for a function’s value by a smaller one while leaving its four derivative coefficients unchanged; 1,540 concern the cubic branch and 1,344 the cosine branch. This is a *conditional composition* result: the original interval must already contain the stated real radicand and its argument must lie in the stated input interval.

Cubic branch. For a rational $B \geq 1$ and $P_B(u) = 1 - u^2 + \frac{B+1}{5}u^3$ on a nonnegative interval $[a, b]$, the only possible interior critical point is $u_* = \frac{10}{3(B+1)}$, so the range is controlled by the two endpoint values together with, if $u_* \in [a, b]$, the additional rational value $P_B(u_*) = 1 - \frac{100}{27(B+1)^2}$. The checker tests that the proposed range contains the endpoint values and either excludes u_* or contains its value; the proof uses compactness to attain the extrema and Fermat’s theorem at an interior one.

Cosine branch. Here the radicand is

$$R_B(u) = 1 - \frac{2A(1 - \cos((B+1)u))}{(B+1)^2}, \quad A = \frac{81}{80(1 - \cos \frac{9}{2})},$$

and $-1 \leq \cos \leq 1$ gives $1 - \frac{4A}{(B+1)^2} \leq R_B(u) \leq 1$ for every real u . A separately checked sine/cosine endpoint certificate at $9/2$ proves $A \leq A_+ := \frac{3591567126}{2^{32}}$, so the rational interval $[1 - \frac{4A_+}{(B+1)^2}, 1]$ contains $R_B(u)$ for every real u ; in particular this stays valid if outward rounding pushes the numerical input slightly past a branch endpoint. The branch conditions used to bound the characteristic function are separate obligations.

Intersection. If $I \ni F(x)$ is the original interval and the analytic argument proves $J \ni F(x)$, then $I \cap J \ni F(x)$, and the checker verifies $I \cap J \subseteq I_{\text{new}}$. Merely checking $I_{\text{new}} \subseteq I$ would *not* prove $F(x) \in I_{\text{new}}$. Only the zeroth enclosure changes; the real function and all derivative enclosures are the same.

LEAN. `BerryEsseen/Numerics/RadicandRefinement.lean` (`Jet.Encloses.refine_radicand`, with the original interval and the real-expression identity as explicit premises) and `BerryEsseen/Certificates/FiniteRadicandRefinements/Complete.lean` (2,884 instances across 29 batch files).

12.7 Cotangent tables

The kernel tables store integer bounds for $q(\pi k/4096)$ (both directions), for $e^{-k/1024}$, and for $\cos(k/4096)$. Their meanings are proved as follows.

- *Exponential entries.* The integer polynomial is exactly the degree-24 Taylor polynomial for $e^{k/1024}$ with cleared denominators; every omitted term is positive, so taking reciprocals gives the required *upper* bound for $e^{-k/1024}$.
- *Cosine entries.* The polynomial has degree 32. On $[0, 7]$, group the omitted terms of the absolutely convergent cosine series into negative–positive pairs starting at degrees 34 and 36; each pair is nonpositive because $x^2 \leq (m+1)(m+2)$ for $m \geq 34$ and $0 \leq x \leq 7$. Hence the degree-32 polynomial is an upper bound throughout the table’s argument range. The alternate branch using the trivial bound $\cos \leq 1$ is also proved.
- *Cotangent entries.* These are exactly (16), with the positivity guards $A_- \geq 0$ and $D_- > 0$ being the ones the integer tables check, plus the conversion from the rational bounds on π to the intended table argument.

LEAN. `BerryEsseen/Numerics/ScalarTableCore.lean` (unchanged executable arithmetic from the saved certificate, at scale 2^{64}), `.../ScalarTables.lean` (`tableEntry_exp`, `tableEntry_cos`, `tableEntry_q_lower`, `tableEntry_q_upper` and their “at a point” variants), `.../CotangentTable.lean`, `BerryEsseen/CotangentPolynomials.lean`, `BerryEsseen/CotangentCorrection.lean`; `BerryEsseen/ScalarRemainders.lean` justifies replacing `exp`, `sin` and `cos` by rational polynomials with explicit error intervals, and `BerryEsseen/ExponentialRemainder.lean` supplies the exponential remainders.

12.8 Inventory of the checked collection

Table 6 gives the counts of the certified objects, each verified against the corresponding generated manifest or by listing the directory.

Remark 12.2 (Recorded rational margins). The module `BerryEsseen/CertificateMargins.lean` records, as rational inequalities only, the following strict comparisons with the target $\frac{293}{625} = 0.4688$:

$$\begin{array}{llll} \text{small_ell_margin} : & \frac{4473}{10000} = 0.4473 & < & \frac{293}{625} \\ \text{large_ell_margin} : & \frac{11}{24} = 0.458\overline{3} & < & \frac{293}{625} \\ \text{all_n_margin} : & \frac{4302436550217957753}{2^{63}} = 0.4664711\dots & < & \frac{293}{625} \\ \text{band_margin} : & \frac{8647540607797249413}{2^{64}} = 0.4687841\dots & < & \frac{293}{625} \\ \text{finite_margin} : & \frac{2011606487}{2^{32}} = 0.4683636\dots & < & \frac{293}{625} \end{array}$$

The module’s own docstring states that these are rational inequalities only and that connecting the numbers to probability errors requires the analytic soundness and complete certificate proofs, so this module alone is not an upper bound for C_{iid} . The proved regional constant for the small-fraction region is the $\frac{183}{400}$ of Theorem 8.1.

Object	Count	Location
Interval cells (scale 2^{64})	467	Certificates/ScalarFullH/
Finite-sample complete cells	174	Certificates/FiniteCompleteCells/
Finite-sample parameter blocks	17	Certificates/FiniteParameterBlocks/
High-frequency Taylor panels	1806	Certificates/FiniteHighTaylorPanels/
Low-frequency Taylor panels	1101	Certificates/FiniteLowTaylorPanels/
Normal-correction Taylor panels	669	Certificates/FiniteNormalPanels/
Endpoint (Darboux) panels	458	Certificates/FiniteDarbouxPanels/
Gaussian tails	174	Certificates/FiniteTails/
Exponential seeds	3514	Certificates/FiniteExpSeeds/
Paired sine/cosine seeds	4248	Certificates/FiniteTrigSeeds/
Radicand refinements	2884	Certificates/FiniteRadicandRefinements/
Scalar constant enclosures	693	Certificates/FiniteScalarConstants/
<i>Illustrative individual panel files (not extra certificates):</i>		
High-frequency example panels	7	Certificates/FiniteHighPanels/
Low-frequency example panels	6	Certificates/FiniteLowPanels/

Table 6: The checked certificate collection. The seven high-frequency example panels — (row, panel) = (0, 0), (0, 1), (0, 2), (0, 7), (0, 23), (166, 1), (173, 30) — cover all six branch/kernel combinations, both cosine signs, and sample-size floor 19; they are imported by the supporting library to illustrate the general construction, while the complete collection lives in the batch files.

12.9 What is not trusted

The Python generators are *outside* the trusted proof. They propose cutoffs, subdivisions, panel counts, rational bounds, seeds and refinement witnesses; Lean then verifies the meaning and correctness of the resulting certificates. Concretely:

- No floating-point result, external interval library, or producer success flag is a premise of any Lean theorem.
- Source hashes and manifests provide reproducibility, not mathematical content; they are `import_manifest.json` and `manifest.json` in the certificate directories, and `finite_scalar_meanings.json` and `finite20_coalesced.jsonl` under `data/`.
- A floating-point maximum, or a collection of positive sampled margins, is never used as a substitute for the interval inequalities.
- The import script for the interval cells preserves the frozen integer records and their arithmetic conditions; it changes module wrappers, removes diagnostic printing, tightens a former 0.4689 budget to 0.4688, and uses the proved `CertifiedCell` type for the final assembly.

Reproduction (from a checkout with Lean and the mathlib cache installed):

```
lake exe cache get
python3 scripts/build_certificates.py          # or --batch-size 2 for CI
```

The batching limits the number of concurrent compiler processes; it does not change any proof check. One build driver per checkout should be used, since overlapping Lake invocations can observe partially written dependency outputs.

13 Assembling the upper bound

Theorem 13.1 (Pointwise universal bound). *Let μ be admissible, $n \geq 1$ and $x \in \mathbb{R}$. Then $N(\mu, n, x) \leq \frac{293}{625}$.*

Proof. Write $\ell = \beta/\sqrt{n}$ and split into four exhaustive cases, which is exactly the case analysis of `BerryEsseen/UniversalUpperBound.lean`.

1. If $\ell \leq \frac{1}{20}$: Theorem 8.1 gives $N \leq \frac{183}{400} \leq \frac{293}{625}$.

Directory	decide +kernel occurrences
ScalarFullH	349,824
FiniteLowTaylorPanels	263,610
FiniteHighTaylorPanels	256,223
FiniteNormalPanels	76,936
FiniteTrigSeeds	4,249
FiniteExpSeeds	3,515
FiniteRadicanRefinements	2,885
FiniteLowPanels	1,276
FiniteHighPanels	973
FiniteScalarConstants	694
FiniteDarbouxPanels	459
FiniteTails	349
FiniteCompleteCells	175
FiniteHighCells	175
FiniteLowCells	175
FiniteNormalCells	175
FiniteParameterBlocks	17
Total	961,710

Table 7: Kernel-checked finite decisions, counted over `BerryEsseen/Certificates/` with `grep -rc "decide +kernel" ... | awk -F: '{s+=$2} END {print s}'`. Every one of these is evaluated by Lean’s kernel on exact arithmetic; none uses `native_decide`.

2. Otherwise $\ell \geq \frac{1}{20}$. If $\beta \geq 2$: Theorem 10.1(a).
3. Otherwise $\beta < 2$. If $n \geq 20$: Theorem 10.1(b).
4. Otherwise $1 \leq n < 20$ and $\beta < 2$: Theorem 11.1.

Each branch of cases 2–4 internally falls back on Corollary 9.2 where ℓ exceeds the last certified endpoint $\frac{6}{5}$. □

Proof of Theorem 1.2. By Lemma 2.4 and Theorem 13.1, $C_{\text{iid}} \leq \frac{293}{625} = 0.4688$. □

The Lean text of the case split is short enough to quote in full:

```

theorem normalizedError_universal_upper_bound {mu : ProbabilityMeasure R}
  (hmu : Admissible mu) {n : Nat} (hn : 0 < n) (x : R) :
  normalizedError mu n x <= 0.4688 := by
  by_cases hs : thirdAbsMoment mu / sqrt n <= 1 / 20
  . exact normalizedError_le_target_of_small_fraction hmu hn hs x
  have hell : 1 / 20 <= thirdAbsMoment mu / sqrt n := (lt_of_not_ge hs).le
  by_cases hb : 2 <= thirdAbsMoment mu
  . exact ScalarFullHAll.large_moment_normalizedError hmu hn hb hell x
  by_cases hn20 : 20 <= n
  . exact ScalarFullHAll.large_sample_normalizedError hmu hn20 hell x
  exact FiniteParameterBlocks.finite_normalizedError hmu hn (by omega)
  (lt_of_not_ge hb) x

theorem berryEsseenConstant_upper_bound : berryEsseenConstant <= 0.4688 := by
  apply (constant_le_iff _).mpr
  intro mu hmu n hn x
  exact normalizedError_universal_upper_bound hmu hn x

```

(Unicode has been transliterated for legibility; the actual file uses the Unicode symbols for $\mathbb{R}$, $\mathbb{N}$, $\leq$ and μ .)

Corollary 13.2. $C_{\text{iid}} < \infty$, and in particular for every centred variance-one law with finite third absolute moment and every $n \geq 1$,

$$\sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)| \leq 0.4688 \frac{\mathbb{E}|X|^3}{\sqrt{n}}.$$

Combined with Theorem 1.1, $0.40 \leq C_{\text{iid}} \leq 0.4688$.

LEAN. `BerryEsseen.normalizedError_universal_upper_bound` and `BerryEsseen.berryEsseenConstant_upper_bound` in `BerryEsseen/UniversalUpperBound.lean`; `Solution.lean` re-exports the two theorems under the comparator’s names and also proves their conjunction `berry_esseen_constant_bounds`.

14 Verification status and trust boundary

14.1 Axioms and tactics

The two theorems are proved with no `sorry` and no custom axiom, and their axiom closure is exactly

`{propext, Classical.choice, Quot.sound},`

the three standard mathlib axioms. In particular `native_decide` does not occur anywhere in the 0.4688 development: a search of `BerryEsseen/` outside `BerryEsseen/Upper0423/`, together with `Challenge.lean` and `Solution.lean`, returns no occurrence of `native_decide` and no occurrence of `sorry`. All 961,710 finite decisions are `decide +kernel`, i.e. kernel evaluation of exact integer and rational arithmetic (Table 7).

This matters for the trust boundary: a `native_decide` check delegates evaluation to compiled code and adds an auxiliary axiom per check, whereas `decide +kernel` is checked by the same kernel that checks the rest of the proof. The sharper 0.423 development in the same repository makes the opposite choice; see Section 1.5.

14.2 The challenge file and the comparator

`Challenge.lean` contains all definitions of Section 2 written out and states the two theorems with intentional proof holes. It imports only mathlib, and only three files of it: `Mathlib.MeasureTheory.Constructions.Pi`, `Mathlib.MeasureTheory.Measure.ProbabilityMeasure` and `Mathlib.Probability.Distributions.Gaussian.Real`. `Solution.lean` proves the same two statements from the supporting library.

`comparator/config.json` is

```
{ "challenge_module": "Challenge",
  "solution_module": "Solution",
  "theorem_names": [ "berry_esseen_constant_lower_bound",
                     "berry_esseen_constant_upper_bound" ],
  "permitted_axioms": [ "propext", "Quot.sound", "Classical.choice" ] }
```

and the pinned tools are those of Table 8. The runner script `scripts/run_comparator_linux.sh` checks out these revisions, builds the tools and invokes the Landrun sandbox under the upstream-recommended `systemd-run` restriction; it does not use the macOS development adapter. The workflow `.github/workflows/comparator.yml` can be dispatched without inputs; it builds the certificates in bounded batches (`--batch-size 2`), asserts `git diff --exit-code` so that no source file was modified during the build, provisions swap for the replay, and runs the comparator, uploading its log as an artifact. An optional `proof-cache-tag` may supply precompiled supporting outputs; the workflow accepts a cache only if its supporting source contents, project configuration and dependency identities match the checkout, it verifies every archive part’s size and SHA-256 before extraction, and it excludes `BerryEsseen/Defs`, `Challenge` and `Solution` so that those are always rebuilt from the current source.

14.3 Status, stated exactly

- **Proved in Lean.** Theorems 1.1 and 1.2, with the axiom closure and tactic discipline above.
- **Comparator: pending.** The final pinned, sandboxed Linux comparator run — statement comparison against `Challenge.lean`, axiom whitelist check, and kernel replay of the dependency closure — has *not* been reported as passing. The repository states this in `README.md` (“the separate pinned,

Tool	Pinned git revision
leanprover/comparator	2312244ac716564a61cc0bf4e107d9abf1757a61
leanprover/lean4export	cacf989bd75f608700820f6afc595f32e7a99a4d
Zouuup/landrun	811cfff51ceaf3d9843708aa6d22e9b84ccac8b4

Table 8: The pinned verification tools [5].

sandboxed Linux comparator check is pending”), in `comparator/README.md` (“The final comparator run is still pending”) and in every proof note. Nothing in this document should be read as a claim of comparator completion. A passing ordinary `lake build` is reported separately from the comparator’s statement comparison and kernel replay; any local macOS run using the upstream development adapter is to be identified separately and is not a sandboxed verification.

- **Local build.** The full library was rebuilt from scratch on the machine used to prepare this document (a ten-core Apple Silicon laptop): `python3 scripts/build_certificates.py --batch-size 4` followed by `lake build Solution Challenge` completed successfully, reporting 5,566 and 6,117 finished jobs, in 183 minutes of wall-clock time (583 minutes of CPU time). On the resulting build, `#print axioms` reports exactly `[propext, Classical.choice, Quot.sound]` for both `berry_esseen_constant_lower_bound` and `berry_esseen_constant_upper_bound`, and their statements are the two displayed in Section 2. This is an ordinary Lean build and axiom query; it is not the sandboxed comparator run, which remains pending.

14.4 Sanity properties of the formalization

Three features are worth checking against the informal statement of the Berry–Esseen problem, because they are the usual places where a formalization can be weaker than it appears.

1. *The class of laws is not restricted.* The supremum in Definition 2.3 ranges over all `ProbabilityMeasures` on $\mathbb{R}$ satisfying Definition 2.1; there is no finiteness, boundedness, or discreteness hypothesis, and no placeholder definition.
2. *Independence is genuine.* μ_n is the pushforward of a finite product measure under the normalized sum, and $\hat{\mu}_n(t) = f(t/\sqrt{n})^n$ is proved, not assumed.
3. *Finiteness is a theorem.* C_{iid} is defined in $[0, \infty]$; the upper bound is what makes it finite.

A Index of Lean files by role

File (under <code>BerryEsseen/</code> unless noted)	Role
<code>../Challenge.lean</code>	self-contained statement with two proof holes; imports only <code>mathlib</code>
<code>../Solution.lean</code>	the two comparator targets and their conjunction
<code>Defs.lean</code>	Definitions 2.1–2.3
<code>Constant.lean</code>	Lemma 2.4
<i>Lower bound</i>	
<code>BernoulliWitness.lean</code>	the witness law, its moments, the 2^6 -term exact probability
<code>GaussianBounds.lean</code>	normal density and CDF estimates, Lemma 3.4
<code>LowerBound.lean</code>	Theorem 1.1
<i>Characteristic-function estimates</i>	
<code>MomentExcess.lean</code>	Lemma 4.1
<code>IntegralQuadratic.lean,</code>	Lemma 4.2
<code>WeightedMoments.lean</code>	
<code>TrigonometricAnchors.lean</code>	Lemma 4.3
<code>ImaginaryPart.lean</code>	Proposition 4.4
<code>RealPart.lean</code>	Proposition 4.5
<code>CharacteristicBounds.lean</code>	coordinate combination, $\beta = 1 \Rightarrow f = \cos$
<code>GaussianCosine.lean,</code>	Lemma 4.7 and its polynomial refinement
<code>GaussianPolynomial.lean</code>	
<code>ClassicalError.lean</code>	the classical third-order error bound
<code>SumBounds.lean</code>	Definition 4.8
<i>Global modulus</i>	

File	Role
Symmetrization.lean	Lemma 5.2
CosineMinorant.lean,	Proposition 5.4
CosineProfile.lean, ConvexSplice.lean,	
FullMinorant.lean	
FullModulus.lean	Theorem 5.1
<i>Prawitz smoothing</i>	
KernelRecurrence.lean	the two-monotone-sequence argument
PrawitzKernel.lean	integrability and the shift recurrence (15)
OscillatoryIntegral.lean	interval form of Riemann–Lebesgue
PrawitzMajorant.lean	Theorem 6.2
PrawitzProbability.lean	Proposition 6.3
PrawitzFubini.lean,	Proposition 6.4
CharacteristicProjection.lean,	
PrawitzFourier.lean	
GaussianInversion.lean,	Lemma 6.5
FirstMomentFourier.lean	
SmoothingIntegrability.lean,	Theorem 6.6
SmoothingSplit.lean,	
SmoothingAlgebra.lean,	
PrawitzSmoothing.lean	
SmoothingMajorants.lean	Corollary 6.7
PrawitzIntegralForm.lean	integral reformulation of the kernel
<i>Compression</i>	
GeometricDamping.lean	A , G_q , Lemma 7.2, (22)
DampedPower.lean, PowerComparison.lean	difference of powers, replacement of $1/n$ by v
ScalarCompression.lean	Proposition 7.3 for the actual i.i.d. law
VectorRemainder.lean,	moment-band coefficients and the vector prefactor
VectorCompression.lean	
ParameterCaps.lean	Lemma 7.4
CompressedSmoothing.lean,	the envelope functional (29) and its integrability
MixedSmoothing.lean	
ParameterCellBounds.lean	Lemma 2.5 and the two cell interfaces
<i>Region I and II</i>	
SmallFractionCoefficients.lean,	Theorem 8.1
SmallFractionPower.lean,	
SmallFractionKernels.lean,	
SmallFractionGaussian.lean,	
SmallFractionLow.lean,	
SmallFractionNormal.lean,	
SmallFractionDecay.lean,	
SmallFractionHigh.lean,	
SmallFractionRemainders.lean,	
SmallFractionBudget.lean,	
SmallFraction.lean	
Cantelli.lean, LargeFraction.lean,	Theorem 9.1, Corollary 9.2
IndependentSums.lean	
<i>Region III</i>	
DecayPanels.lean, DecayWitnesses.lean	Lemma 10.2
SmoothingPanels.lean,	Propositions 10.3–10.5
LowSmoothingPanels.lean,	
VectorSmoothingPanels.lean,	
PanelSums.lean, GaussianTailBound.lean	
Numerics/ScalarPanelCore.lean,	executable record formats and validators
Numerics/FixedCellCore.lean,	
Numerics/GeneralCellCore.lean,	
Numerics/UpperPanelCore.lean	
Numerics/Cell*Facts.lean,	record-to-integral soundness (Table 3) and Theorem 10.7
Numerics/CellApplication.lean,	
Numerics/CellPanelSums.lean,	
Numerics/CellCertificate.lean,	
Numerics/CertifiedCell*.lean,	
Numerics/HighGrid.lean,	
Numerics/IndexedPanelLists.lean	
Certificates/ScalarFullH/	the 467 cells, the tables, the coverings
<i>Region IV</i>	
FiniteSampleBounds.lean,	Theorem 11.2
FiniteModulus.lean, FiniteKernels.lean,	
FiniteGeometricAverage.lean,	
FiniteSmoothing.lean	

File	Role
<code>FiniteRadicandRange.lean</code> , <code>FiniteDarboux.lean</code>	Lemma 11.4, Proposition 11.5
<code>FiniteLowPanels.lean</code> , <code>FiniteHighPanels.lean</code> , <code>FiniteNormalPanels.lean</code>	generic real-integral comparisons per panel kind
<code>FiniteLowCover.lean</code> , <code>FiniteHighCover.lean</code> , <code>FiniteNormalCover.lean</code>	adjacency and completeness of each panel list
<code>FiniteCellCertificate.lean</code> , <code>FiniteParameterCover.lean</code> , <code>FiniteMomentCap.lean</code>	Definition 11.6, Theorem 11.1
<code>Certificates/Finite*</code>	the checked collection of Table 6
<i>Numerics infrastructure</i> <code>TaylorPanel.lean</code> , <code>Numerics/TaylorPanelSoundness.lean</code>	Theorem 11.3
<code>Numerics/JetCore.lean</code> , <code>Numerics/IntervalSoundness.lean</code>	Definition 12.1 and outward rounding
<code>Numerics/*Jet.lean</code> , <code>Numerics/EnclosesOn.lean</code> , <code>Numerics/JetSoundness.lean</code> , <code>Numerics/JetInputs.lean</code> , <code>Numerics/JetPanelRange.lean</code>	jet recurrences and their derivative semantics
<code>Numerics/ScalarEnclosure.lean</code>	exact-value scalar enclosures
<code>Numerics/ExpSeed*.lean</code> , <code>Numerics/TrigSeed*.lean</code> , <code>Numerics/Trig*Soundness.lean</code> , <code>TrigonometricRange.lean</code>	Sections 12.4, 12.5
<code>Numerics/RadicandRefinement.lean</code>	radicand refinements
<code>Numerics/ScalarTable*.lean</code> , <code>Numerics/CotangentTable.lean</code> , <code>CotangentPolynomials.lean</code> , <code>CotangentCorrection.lean</code> , <code>ScalarRemainders.lean</code> , <code>ExponentialRemainder.lean</code>	table semantics and rational remainders
<code>Numerics/DarbouxCertificate.lean</code> , <code>Numerics/FiniteTailCertificate.lean</code> , <code>Numerics/GeometricWitness.lean</code> , <code>Numerics/DecayCertificate.lean</code>	the remaining per-record checkers

B The four smoothing terms at a glance

For orientation, Table 10 lists which estimate controls which of the four terms of Theorem 6.6 in each region.

Term	Region I	Region III	Region IV
low-frequency error	(26) + (28)	$\mathcal{E}$ via (21)	$\mathcal{L}$ via (25)
high-frequency magnitude	Theorem 5.1 + Lemma 8.9	$e^{-\Psi}$ via Theorem 5.1	M_B^N , line/cosine/anchor branches
normal correction	(17)	Proposition 10.3(b)	K_C via (16)
Gaussian tail	elementary, $T \geq 60$	Proposition 10.5	Section 11.6

Table 10: Which estimate controls which term. Region II (Cantelli) uses no smoothing at all.

Acknowledgement of scope. This document is a mathematical exposition of a formal development. It reports the development’s own recorded status and provenance (Sections 1.4, 1.6, 14) and makes no claim of priority, and no claim about the correctness of any other development, in particular not of [16].

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